

Measurement Properties of Dictionary-Based ESG-Washing Indices:

A Technical Reassessment and Longitudinal Extension

1 Scope and relationship to the original study

This study is an adaptation of the ESG-washing index of Lagasio (2024), not a strict replication of it. We re-implement the index architecture and examine its measurement properties, but three inputs differ by design — the corpus, the ESG vocabulary and the sentiment dictionary — and one of the three cannot be aligned at all from the published record, because the original never publishes its vocabulary or its size. The delta is stated here, at the outset, so that no later result is read as a failed reproduction of something it never attempted to reproduce.

What is carried over. Two things. First, the index architecture: a document-level tone score and a document-level substance score, each rescaled across the sample, differenced, and thresholded at zero to produce a binary label. Second, the tone-minus-substance logic that motivates it — positive language in excess of measurable substantive content as the observable signature of ESG-washing. Both are carried over unchanged, including the zero threshold and the labelling rule, which the original states explicitly (§6). Where the original’s specification is incomplete we do not close the gap silently: §2 documents each silence, and §6 carries three substance specifications precisely because the original’s description does not determine one.

Where we diverge, and why. Table 1 sets out the full delta. Three of its entries are input choices that require a reason; a fourth is a pipeline stage with no counterpart in the original.

Corpus. The original starts from the top 2,000 firms by market capitalisation in Refinitiv and retains the 749 with a standalone, English-language ESG report for 2023; “companies that integrated their ESG disclosure within their annual financial reports were excluded” (Lagasio, 2024, p. 3). Our corpus is drawn largely from the population that rule excludes: 343 unique reports from 49 firms in seven European countries over 2018–2024, of which 235 are annual reports, 101 universal registration documents (URD) and 7 are standalone sustainability reports. The entailment runs from the panel design, and we state its direction rather than present the divergence as unavoidable: we adopt a balanced seven-year panel — longitudinal work being what the original’s own conclusion calls for — and it is that choice which rules out the original’s inclusion criterion. A balanced panel requires one annually recurring disclosure per firm in every year of the window. The annual report, and in France the URD, is a periodic regulatory filing that every firm in the frame publishes in every year; a standalone sustainability report is issued at the firm’s discretion, so neither its existence nor its recurrence over seven consecutive years can be assumed. Our own corpus shows how binding that is here: of the 49 firms, exactly one — Sanofi-Aventis — is represented by standalone sustainability reports across the window, and it is so in all seven years, so a panel restricted to that document type would have contained one firm. What our records hold is the document *selected* for each firm-year, not what each firm made available, so this is a statement about what a document-type restriction would have left us with and not about what these firms published. Relatedly, 43 of the 49 firms are represented

by a single document type across the whole window; the six that change do so once, from annual report to URD, consistent with the French regime. Retaining the original’s rule would have made panel membership a function of the firm’s chosen reporting vehicle — a disclosure behaviour in its own right — and left the panel unbalanced. The restriction to 49 firms in seven European countries is a separate sampling-frame decision and not a consequence of the panel design; it, and the operational consequence of mixing document types inside a single index, are treated in §3 and §12.

Lexicon. The original’s keyword list is attributed to GRI (2021) and SASB (2018), but its size is never stated, the list is never published, and the data availability statement reads in full “No” (§2). We therefore built a purpose-designed 154-term vocabulary and publish it in full (§A). The consequence is stated plainly rather than deferred to a limitations paragraph: our substance score and the original’s are not the same measurement, and no numerical difference between the two indices can be attributed cleanly to any single design choice (§5).

Sentiment dictionary. The original scores tone with TextBlob polarity, a general-purpose pattern-based tool. We use the Loughran–McDonald finance dictionary (Loughran and McDonald, 2011), which is calibrated on disclosure language; the original’s own conclusion names “refining sentiment analysis techniques” (Lagasio, 2024, p. 10) among the extensions it anticipates. The two are not interchangeable instruments, and each carries its own defects — ours are set out in §12, and what the original discloses about its own instrument is audited in §2 — so the tone components are no more directly comparable across the two studies than the substance components are.

Text unit. A fourth divergence is a stage rather than an input: we vectorise lexically extracted ESG zones, not whole reports (§4). It has no counterpart in the original, which vectorises the report entire; it is required here because ESG disclosure in an annual report or URD sits inside a much larger financial and legal document; and it fixes the denominator against which the length effects of §7 operate. Its cost — an untested filtering stage upstream of every number we report — is conceded in §12.

Temporal extension. The original states that a panel was unnecessary for its purpose — “a cross-sectional dataset focused on a single year (2023) is sufficient. Our methodology does not require a panel dataset spanning multiple years” (Lagasio, 2024, p. 3) — and its conclusion states the opposite: “The cross-sectional nature of our data limits causal inferences, highlighting the need for longitudinal studies to examine how changes in corporate characteristics and regulatory environments affect ESGSI over time” (Lagasio, 2024, p. 10). The two sit in tension in the original; our design resolves it in favour of the second. §11 reports the resulting series, conditional on the measurement findings established before it.

What the technical sections establish. §2 audits what the original specifies and what it leaves unspecified; §3–§6 document our corpus, extraction, lexicon and index. §7 shows that *vector* normalisation, not term weighting, governs what a TF-IDF substance score measures, and §8 that the original’s headline conclusion turns on two undisclosed choices, one of them a *score* normalisation. §9 and §10 establish which substance specification survives an independent validity check and how far the labels move across specifications. §11 gives the 2018–2024 series, §12 the limitations, and §A the full parameter, vocabulary and lexicon listings.

2 What the original specifies, and what it does not

This section sets out the index of Lagasio (2024) as published, then delimits where the published specification stops. The exercise is documentary: it uses none of our pipeline and asserts nothing about what the original’s code did. Where the article is silent we record the silence; we do not substitute what standard practice would have been. There is one exception, and because it is the

Table 1: Specification delta against Lagasio (2024).

	Lagasio (2024)	This study
Design	Cross-sectional, 2023 only	Balanced panel, 2018–2024
Documents	749 standalone ESG reports, one per firm	343 unique reports: 235 annual reports, 101 URD, 7 standalone sustainability reports
Firms	749, worldwide, from a Refinitiv top-2,000 starting sample	49, seven European countries, each present in all seven years
Inclusion rule	English only; firms reporting ESG inside the annual report excluded	One annually recurring filing per firm-year, chiefly the annual report or URD — largely the population the original excludes
Text vectorised	Whole report	Lexically extracted ESG zones (§4)
ESG vocabulary	GRI/SASB-derived; size not stated, list not published	154 terms, published in full (§A)
Tone dictionary	TextBlob polarity	Loughran–McDonald
Substance score	One score; aggregation over the vocabulary not specified	Three specifications, reported side by side (§6)
Score normalisation	Prose says min–max; printed formula is a z -score difference	z -score; both examined (§8)

pivot on which a referee would press we make it in the open rather than in a footnote: the `norm` setting is inferable from the convention that only non-default library arguments are reported, we make that inference and rely on it in §7, and we accordingly do not count it among the gaps that make this index irreproducible. The article is open access under CC BY 4.0, so contested points are quoted rather than paraphrased. Page numbers are the article’s own.

2.1 The published construction

The index has two components. The sentiment component SEN_i is the score returned by the TextBlob library for report i (p. 4), which the index definition calls the “sentiment polarity score”. The substance component SUS_i is derived from scikit-learn’s `TfidfVectorizer`, version 0.24.2 (p. 3), evaluated against an author-constructed list of ESG keywords. The index is the difference of the two components after score normalisation, and the labelling rule is definitional: $ESGSI > 0$ is “Potential ESGwashing”, $ESGSI \leq 0$ is “Likely Genuine” (pp. 4–5). No calibration against an external benchmark is reported.

The vectoriser settings are disclosed in two sentences (p. 4):

“We use Scikit-learn’s `TfidfVectorizer` with settings chosen to balance comprehensiveness and computational efficiency, in line with best practices in NLP research. These settings include ignoring terms that appear in more than 95 % of documents, only considering terms that appear in at least 2 documents, limiting the vocabulary to the top 10,000 terms, and considering both unigrams and bigrams.”

In library terms, `max_df=0.95`, `min_df=2`, `max_features=10000`, `ngram_range=(1,2)`. These four are the only vectoriser settings the article discloses, and they are described in prose only: the article prints no parameter names, no code listing and no equation for SUS_i . Two further sentences in the same paragraph bear on the vectoriser and add no settings. The first establishes that IDF weighting is in use: the method evaluates “the relative importance of each term within the documents against the entire dataset” (p. 4). The second states that the vectoriser “is used with adjustable settings to accommodate the unique features of different sustainability reports” (p. 4); no setting beyond the four above is given.

The step from per-term values to a document-level score is where the specification stops. Two sentences bear on it. From the original’s §3.2.1 (p. 4): “Each document is then vectorized with the TF-IDF model to calculate a score for each term based on its frequency relative to other documents.” From its §3.2.3 (p. 4): “It combines sentiment scores with TF-IDF scores of sustainability terms to objectively evaluate how extensively sustainability topics are discussed in the reports.” The abstract adds only “integrating sentiment analysis with the frequency of sustainability terms to calculate the ESGSI” (p. 1). In the index formula, SUS_i is glossed as “sustainability score for company i” — a name, not a construction.

2.2 What is disclosed, and what is not

Table 2 separates the two.

Table 2: Elements of the original specification, by disclosure status.

Element	Status in Lagasio (2024)
Vectoriser implementation and version	Disclosed: <code>scikit-learn TfidfVectorizer</code> 0.24.2 (pp. 3–4)
<code>max_df</code> , <code>min_df</code> , <code>max_features</code> , <code>ngram_range</code>	Disclosed in prose (p. 4)
IDF weighting in use	Disclosed (p. 4)
Sentiment instrument	Disclosed: <code>TextBlob</code> polarity (p. 4)
Keyword-list provenance	Disclosed: GRI (2021) and SASB (2018) standards (p. 4)
Labelling threshold	Disclosed, definitional (pp. 4–5)
<code>norm</code> (vector normalisation)	Not stated; library default reasonably inferable
<code>sublinear_tf</code> , <code>smooth_idf</code>	Not stated; library defaults reasonably inferable
Aggregation from per-term TF-IDF to SUS_i	Not stated; not a library setting, so no default supplies it
Whether TF-IDF is computed over the 10,000-term vocabulary and then subset to the keywords, or the vectoriser is restricted to the keywords	Not stated
Keyword-list size and contents	Not published; three example terms only
Custom stopword additions to the NLTK defaults	Not published
Whether raw or preprocessed text reached <code>TextBlob</code>	Not stated
Score normalisation (min–max or <i>z</i> -score)	Contradictory; see §8
Share of firms flagged “Potential ESGwashing”	Never reported
Cohen’s Kappa, 5-fold cross-validation, TF-IDF sensitivity analysis	Announced (p. 5), never reported
Data availability statement	“No” (p. 10)

Four entries warrant expansion. First, *vector* normalisation. The `norm` parameter is never named; the strings *L2* and *Euclidean* do not occur in the article at all; and no rescaling of document vectors is described under any other wording. The article does use the word *normalization*, but of two other operations: lowercasing and character stripping during preprocessing (p. 3), and the rescaling of the finished component scores across the sample (p. 4), which is a different operation at a different stage (§8). Table 2 therefore records `norm` as not stated, which is a fact about the text; it does not record it as a reproducibility gap, and we concede the point openly. The four disclosed settings are exactly four non-default `TfidfVectorizer` arguments, so a reader following the ordinary convention that only departures from library defaults are reported would infer that the remaining defaults — `norm='l2'` among them — were in force. That inference is

reasonable, we rely on it ourselves, and §7 draws it explicitly and labels it as an inference. It remains an inference rather than a statement of the article, so we do not say that the original L2-normalised its document vectors and we do not say that it did not; but a silence that a standard convention closes is not where the reproducibility charge of §2.4 rests.

Second, the vocabulary. Its provenance is given — the indicator lists “are based on established ESG reporting frameworks such as Global Reporting Initiative (GRI) Standards ... and Sustainable Accounting Standards Board (SASB) Standards” (p. 4) — but its size is never stated and the list is never published. Three example terms appear (“Diversity”, “Carbon Emissions”, “Equal Employment Opportunities”); there is no appendix, table or supplementary file.

Third, the announced validation (p. 5):

“For the manual data collection process, we employ two independent raters and calculate Cohen’s Kappa to assess agreement ... We use 5-fold cross-validation to evaluate the robustness of our LDA and sentiment analysis models ... We also conduct sensitivity analyses by varying key parameters (e.g., number of LDA topics, TF-IDF settings) to test the robustness of our results.”

Neither a Kappa value nor a cross-validation figure nor a sensitivity result appears anywhere in the article: outside the reference list, the passage above is the only occurrence of *Kappa*, of *cross-validation* and of *sensitivity* in the text. The announced sensitivity analysis names TF-IDF settings explicitly, so a reported version of it would bear directly on the vectoriser silences recorded above (§7).

Fourth, the data availability statement, which consists of the heading “Data availability” and the single word “No” (p. 10).

2.3 One further disagreement, resolved elsewhere

Within its §3.3 the article prints a formula that is a difference of z -scores — reproduced verbatim in §6 — and states in prose, in the same subsection, that “both sentiment and sustainability scores are normalized to a uniform scale using min-max normalization” (p. 4). These are different estimators and the article never reconciles them. We record the disagreement here and adjudicate it in §8, on evidence of a different kind; nothing in the present section turns on which reading is correct.

2.4 Consequence for what follows

A reader who set out to compute this index would have to supply, from outside the article, at minimum: the aggregation rule from per-term values to SUS_i , the keyword list and its size, the score-normalisation rule, and the text passed to the sentiment instrument. Each is a free parameter, and §10 shows that admissible settings differ in the numbers they produce. The index as published is therefore not reproducible from its own description. The obstacle is not access — the full text was available to us — but the completeness of the description.

The first of those four is the strongest instance and carries the charge on its own. The step from a vector of per-term TF-IDF values to a single SUS_i is a modelling decision, not a library argument: sum, mean, maximum and weighted variants are all available, the article names none of them, and no reporting convention supplies a default because the choice is not a default of anything. This is the respect in which it differs from *norm*, where the convention that only non-default arguments are reported does close the gap and we have said so above. The second is of the same kind: a keyword list that is neither published nor sized cannot be reconstructed by any convention, and it is what makes strict replication impossible at the lexicon stage (§5). The third is not a silence at all but a contradiction — the prose of the original’s §3.3 and the formula printed in the same subsection specify different estimators, and no convention adjudicates between two

statements the article itself makes. None of these three is answerable from outside the article, and the charge stands on them.

This governs the design of what follows. Because the original specification does not determine a unique implementation, we do not implement one: we implement several and report results across them. That is why §6 carries three substance specifications rather than one, and why §10 reports how the index and its labels move between them.

3 Data and corpus construction

The corpus comprises 343 unique corporate reports from 49 companies headquartered in seven European countries (Belgium, Finland, France, Germany, Italy, the Netherlands, Spain), one report per company per year from 2018 to 2024. The panel is exactly balanced: every year contributes 49 documents, and at a finer grain every country contributes a fixed number of documents in every year of the panel (France 17, Germany 14, the Netherlands 6, Italy 5, Spain 4, Finland 2, Belgium 1). Balance here is temporal, not cross-national: countries are represented in wildly unequal numbers, as Table 3 shows, but each country’s own count is identical in every one of the seven years, so no year departs compositionally from any other.

Three document types are represented: annual reports, the Universal Registration Document (URD; French *document d’enregistrement universel*), and standalone sustainability reports. Document type was assigned per report from an external metadata file (`muestras_informes.xlsx`), joined to each PDF by country, publication year and a normalised company name — Unicode-folded, punctuation-stripped, and reconciled through a manual company-name map covering divergences between the directory tree and the metadata source. The join resolved all 343 documents; no case was left unmatched.

3.1 The country/type confound is a structural property of the corpus

Table 3 reports the corpus by country and document type. Document type is confounded with country almost completely: 99 of the corpus’s 101 URD filings are French, and all 98 German documents are annual reports. Both are properties of national reporting practice rather than sampling accidents — the URD is the standard annual regulatory filing for French listed companies and routinely carries the extended sustainability disclosure, whereas the German issuers in this sample report ESG content inside the ordinary annual report. We state this here as a design property of the corpus, not as a limitation discovered later: within the pooled panel, a country effect and a document-type effect are not separately identifiable. Because Germany is simultaneously a single country and a single document type (98 of 98 documents are German annual reports), it is the one slice of the panel where a temporal estimate is free of this confound, and §11 reports a within-Germany regression for that reason.

Table 3: Corpus composition by country and document type.

Country	Companies	Annual report	URD	Sustainability report	Total
France	17	13	99	7	119
Germany	14	98	0	0	98
Netherlands	6	40	2	0	42
Italy	5	35	0	0	35
Spain	4	28	0	0	28
Finland	2	14	0	0	14
Belgium	1	7	0	0	7
Total	49	235	101	7	343

Table 4 reports the same corpus by year. Composition by document type is stable across the panel (annual reports 33–36 per year, URD 12–15, sustainability reports 1 per year), which matters when the temporal series (§11) is read against a fixed compositional backdrop rather than a shifting one.

Table 4: Corpus composition by year.

Year	Annual report	URD	Sustainability report	Total
2018	36	12	1	49
2019	33	15	1	49
2020	33	15	1	49
2021	33	15	1	49
2022	34	14	1	49
2023	33	15	1	49
2024	33	15	1	49
Total	235	101	7	343

3.2 Comparability across document types

The panel spans three document types with different structure and different length: annual reports in which ESG disclosure is one section among many financial and operational ones, URD filings that combine regulatory financial, governance and non-financial statements, and standalone sustainability reports that are ESG content throughout. Raw comparability across types cannot be assumed. Every measurement from §4 onward accordingly operates not on whole documents but on the ESG-relevant zones the lexical filter identifies within each one, which is what lets a single index construction (§6) be applied uniformly across all three types. §1 sets out how this corpus relates to the source study’s; here it is the object of measurement.

3.3 Deduplication

Corpus assembly began from 344 collected PDF filings, one more than the final corpus size. A content-hash comparison identified a single duplicate pair: a 2019 annual report for Wolters Kluwer (Netherlands) and a `_repaired` re-issue of the same filing under a different filename. The hash is taken over the preprocessed representation of §5 rather than the raw extraction, so the criterion is that two documents are indistinguishable to every downstream measure; here the two files are in fact identical at the extraction stage as well. The `_repaired` copy was removed as a data-quality control, leaving 343 unique reports. The balanced panel reported above — 49 documents in every year, and a fixed per-country count within every year — holds only after this step: before deduplication, 2019 carried 50 documents, with the Netherlands contributing 7 rather than 6, while every other year already stood at 49.

4 Document extraction: the lexical ESG-zone filter

The original study vectorises each report in its entirety: a standalone sustainability report is, by construction, already ESG text, so Lagasio (2024) needs no step to separate ESG content from anything else. Our corpus is different by design (§1, §3): annual reports and URDs in which ESG disclosure is embedded inside a much larger financial and legal document. Feeding the whole report to the downstream measures would let financial-statement bulk and document length dominate every score. Extraction is therefore a required stage with no counterpart in the original, and it fixes the denominator against which the length effects of §7 and §8 operate.

4.1 Paragraph segmentation

Each PDF is read with PyMuPDF’s block-level extractor (`page.get_text("blocks")`), text blocks only, sorted top-to-bottom, left-to-right, by `src/lexical_document_filter.py` (`LexicalDocumentFilter`) whose full parameter and dependency listing is in §A. A block is accepted as a paragraph only after passing four filters, in order: (i) *margin strip* — dropped if entirely in the top or bottom margin band ($y_1 < 0.08h$ or $y_0 > 0.92h$, h = page height); (ii) *running header/footer suppression* — every margin-band block of at most 120 characters is normalised (digits $\rightarrow$ #) and counted across the document’s pages, and any normalised string recurring on more than 30% of pages is blacklisted and dropped wherever it occurs; (iii) *financial-table detection* — a block is dropped as tabular only if it contains *no* ESG keyword (the regex defined below) *and* either its digit-to-character ratio exceeds 0.35 with at most 6 lines, or every line is under 60 characters, there are at least two lines, and the first has no terminal punctuation (period, exclamation mark, question mark, or colon); the keyword veto keeps a table row that happens to carry ESG content rather than discarding it as noise; (iv) *section headers* — an all-caps block (ignoring non-letters), at least two letters, at most 10 tokens, under 60% single-character tokens, with no token matching a bare-filename pattern (a run of word characters, a period, then a 2–6 character suffix, e.g. `report.pdf`), is emitted as a § TITLE § marker that tags subsequent paragraphs with their section, rather than entering the paragraph stream. Surviving text has URLs stripped, hyphenated line breaks rejoined, non-ASCII characters dropped, and whitespace collapsed.

4.2 Density scoring and zone construction

Each paragraph p is scored by keyword density,

$$\text{density}(p) = 100 \times \frac{\text{kw_count}(p)}{\text{words}(p)}, \quad (1)$$

where `kw_count(p)` counts matches, on the raw paragraph text, of a single compiled regex combining the 154-term vocabulary of §5 — hyphen/space-interchangeable, pluralisable, sorted longest-first so that *carbon neutrality* is not shadowed by *carbon* — with 24 additional patterns not expressible as flat terms: morphological families (`sustainab\w+`, ...), digit-bearing terms (`co2e?`, `scope [123]`, ISO codes), and collocations that disambiguate polysemous heads (*social*, *water*, *energy*) from their non-ESG senses.

A paragraph is *hot* if $\text{density}(p) \geq 1.0$. Every hot paragraph is expanded to a window of one paragraph either side; overlapping or adjacent windows ($\text{lo} \leq \text{prev_hi} + 1$) are merged into one contiguous zone; merged zones under 150 characters are dropped. Table statistics (keyword count, word count, density) are recomputed on each surviving zone’s concatenated text, not summed from its paragraphs. Table 5 lists the free parameters governing hot-paragraph detection and zone construction, none tuned by grid search; the thresholds internal to financial-table and section-header detection above are additional constants, given in prose rather than tabulated. Per PDF, zones are sorted by start offset and joined into one `relevant_text` field, which is all that every later stage (§5 onward) reads.

Table 5: Extraction parameters.

Parameter	Value
Keyword-density threshold (hot paragraph)	1.0 / 100 words
Context window	± 1 paragraph
Minimum zone length	150 characters
Header/footer repeat threshold	$> 30\%$ of pages
Margin bands	top 8% / bottom 8% of page height

4.3 Yield

Table 6 reports extraction volume over all 344 processed PDFs. Extraction runs before deduplication, so its basis is 344 rather than the 343 of the analysed corpus; the extra file is the duplicate removed at §3, and every figure downstream of preprocessing is on 343. On a fixed, disclosed 20-PDF draw (seed 7; §A), not part of the pipeline itself, a mean 35.5% (median 35.7%, range 9.0%–59.6%) of a report’s word count survives into its ESG zones.

Table 6: Extraction yield, aggregated over 344 PDFs.

Statistic	Value
Total zones	74,004
Zones per document (mean, sd)	215.1, 152.9
Extracted words per document (mean)	76,743
Total extracted words	26,399,487
Words per zone (mean across documents)	396.3
Corpus-wide keyword density (extracted text)	3.658 / 100 words
Yield vs. full PDF (20-doc sample, mean / median)	35.5% / 35.7%

4.4 Design rationale

The filter is a pure function of the PDF and the lexicon files: no network call, no external API, no random seed on this path. This matters directly for the determinism claimed in §A. An earlier version of the pipeline extracted fixed-size chunks (1,200 characters, 300-character overlap) and selected among them with a semantic filter — cosine similarity, in an embedding space from a hosted model (Vertex AI `text-embedding-004`), against hand-written anchor queries. That filter was removed and replaced by the deterministic lexical filter above, together with every remaining dependency on the hosted embedding service; the repository records what changed but not why, so no claim is made here about the original motivation. The methodological case for the replacement stands independently of that history: a filter behind a paid, versioned, externally hosted model cannot satisfy a full-determinism claim, because the model can change or be deprecated independently of this codebase and the same PDF is not guaranteed to yield the same selection on a later run.

4.5 Recall is bounded by the lexicon

Extraction’s matching set is a strict superset of the measurement vocabulary: the 24 extra patterns above — morphological families, digit-bearing terms, four precision collocations for polysemous heads, and *paris agreement* and *just transition*, the two phrases §5 records as removed by hand from the 154-term list because they collapse under lemmatisation — can make a paragraph hot and pull a zone in, but are not part of the fixed 154-term count that SUS itself tallies. Two bounds follow, not one: a term outside extraction’s combined set triggers no hot paragraph and enters no zone, so it is invisible everywhere; a term inside that set but outside the 154 — *paris agreement* is the documented case — can still reach a zone without ever being counted by SUS, so its own recall is bounded by the narrower list even for text the filter has already admitted. §12 develops the consequences; it is noted here as a direct property of this step.

5 Preprocessing and lexicon architecture

Every zone identified in §4 passes through a single `TextProcessor`: spaCy’s `en_core_web_md` model, with named-entity recognition and dependency parsing disabled. URLs are stripped,

text is lowercased, tokenised and lemmatised. A token survives only if it is not a stopword, not punctuation, not whitespace, longer than two characters, its lemma is not one of 189 corpus-specific stopwords, and it passes a content-token test: alphanumeric, with at least two alphabetic characters. That last condition replaced an earlier, purely-alphabetic filter that had been silently deleting `co2`, `co2e` and `sf6`; it still excludes bare numerals (2023, 14001) and page references (p12, q1), so digits survive only when attached to letters. Two representations are carried forward: a lemmatised, stopword-filtered token string (`clean_text`), feeding SUS, HEDGE and BREADTH; and a whitespace-collapsed copy of the extracted text with digits and case intact (`raw_text`), the sole input to QUANT.

5.1 Vocabulary and corpus in one representation, by construction

The vocabulary that scores SUS and the lexicon that scores HEDGE are generated artefacts, not hand-written lists — the generation step is the methodological point of this section. Each is authored once, in natural surface form, in a single source file; a build script (`scripts/build_lexicons.py`) passes every entry through the identical `TextProcessor` instance applied to the corpus, so that vocabulary and text are guaranteed to share a representation. The alternative — a second, hand-lemmatised list kept alongside the raw one — invites drift: when the corpus pipeline changes, a hand-maintained list stops tracking it silently. A term whose surface form does not survive preprocessing becomes a dictionary entry that can never match anything, and it fails without warning, because a zero count is indistinguishable from the term’s genuine absence.

Two checks in the build step guard against this. A staleness check compares each source file’s modification time against its generated counterpart, raising an error if the generated file is missing and a warning if the source has since been edited. A collapse check flags any multi-word term that reduces to a single word once preprocessed, rather than adding it silently. The documented case is *paris agreement*, which loses *paris* to the stopword filter and lemmatises to the bare unigram *agreement* — a token occurring 15,459 times in the corpus in senses unrelated to the climate accord (refinancing agreements, employment terminations, sponsorship deals).

The check is advisory, and the distinction matters in a paper about undisclosed methodological choices, so we state it exactly rather than advertise a safeguard the code does not provide. The build script emits a warning naming each collapsed term and then adds it to the generated vocabulary regardless; nothing is withheld automatically. Removal is a separate human decision taken on reading the warning, and it was taken twice: *paris agreement* and *just transition* were deleted by hand from `esg_terms.txt` and re-entered as literal patterns matched against raw text at the extraction stage (§4), which is why *agreement* and *transition* are not bare vocabulary entries. A third collapse was allowed to stand: *due diligence* lemmatises to *diligence*, which is one of the 154 entries, on the judgement that its 4,330 occurrences in the corpus derive almost entirely from the source phrase. An automatic rule would have discarded that one too. What the build step guarantees is that no collapse passes unnoticed; what it does not guarantee is that a collapsed term stays out of the vocabulary.

5.2 The ESG vocabulary

`esg_terms.txt` is 154 canonical terms in natural singular form, organised into seven thematic blocks (Table 7). Run through the build step, the 154 terms yield 154 distinct lemmatised entries with no collisions: 85 unigrams, 62 bigrams, 7 trigrams. The resulting maximum n-gram length, 3, is derived at runtime and supplied as the vectoriser’s `ngram_range` upper bound; scikit-learn’s unigrams-only default would otherwise score every multi-word entry — 45% of the list — as zero, with no error raised.

Because (Lagasio, 2024) never states the size of its GRI/SASB-derived keyword list and publishes only three illustrative terms (§2), our vocabulary cannot be checked against the original’s term for term. This is where strict replication becomes impossible at the lexicon stage, not

Table 7: The 154-term ESG vocabulary, by thematic block.

Thematic block	Terms
Environment: climate and emissions	14
Environment: energy	18
Environment: resources and nature	17
Sustainability and circular economy	12
Social: people and equity	34
Governance and ethics	25
Reporting frameworks and ratings	34
Total	154

merely inconvenient: any divergence between our ESGSI and the original’s cannot be attributed to a single design choice, since the vocabulary is one of several elements differing at once and the original’s is unknown.

5.3 The hedge lexicon

The HEDGE lexicon is built the same way, from a different source. `RAW_LM_dictionary.csv` supplies the Loughran–McDonald word lists (Loughran and McDonald, 2011), 9,752 rows across seven categories; we keep the four bearing on hedging and evasive language — Uncertainty, WeakModal, StrongModal, Constraining, the category names as the source file spells them — 1,245 rows, 1,196 unique lowercase forms. Loughran–McDonald forms are inflected, and crossing them directly against a lemmatised corpus recovered only about 46% of entries. The generated file is instead the union of each entry’s original form and its spaCy lemma, raising the usable lexicon to 1,263 forms, 67 more than the 1,196 that entered lemmatisation.

5.4 External audit of vocabulary coverage

The vocabulary’s coverage was independently audited on 880 zones from four FY2023 reports (Stellantis, adidas, Danone, Enel). The auditor searched the zone text with an extended reference lexicon — GRI, SASB, TCFD and ESRS terminology plus prior literature — by word-boundary regex, excluding any candidate already in our vocabulary. It identified 147 ESG terms present in the zone text and absent from ours, 2,970 occurrences in total; Table 8 lists the four largest. Because matching was lexical, with no paraphrase or synonym detection, and the audit covers only text the extraction stage had already judged ESG-relevant, 147 is a measured lower bound on the vocabulary’s recall, not a full accounting of what it misses.

Table 8: Largest ESG terms omitted from the vocabulary, external audit.

Term	Category	Occurrences
compliance	Governance	436
social	Social	275
supply chain	Governance	241
raw material	Environment	136

Two readings of the shortfall point in opposite directions, and the one that flatters us is wrong. By term count it looks sector-specific: 64 of the 147 appear in only one of the four audited firms. Weighted by occurrence the impression reverses. Those 64 single-firm terms account for 450 of the 2,970 occurrences (15%), while the 22 terms present in *all four* firms — including every row of Table 8 — account for 1,573 (53%). The gap is dominated by general ESG vocabulary

missing everywhere the vocabulary was built to look, not by sector idiosyncrasy, so single-firm concentration is not a basis for discounting it.

The sector concern was in any case an artefact of the sample that measured it. The 147 collapse to 143 distinct candidate strings once near-duplicate surface forms are merged, and every one of them has since been re-measured over all 343 reports — occurrences, documents, companies, countries, and the share held by the largest single company. On that base 103 are broadly dispersed — 102 of which reached the vocabulary, *power purchase agreement* having been lost to a generation-script bug after the audit rejected the bare *ppa* in its favour — 18 are frequent but polysemous, and 22 are concentrated enough in one industry to be called sectoral, carrying 2.0% of the candidates’ occurrence mass. Four companies were too few to tell a sectoral term from a common one. The dispersed and the polysemous classes were incorporated into an expanded 289-entry vocabulary; the 22 sectoral terms were not, a choice §12 records together with its measured effect. The specification reported throughout this paper remains the 154 terms of Table 7, and the expansion is a robustness check on it (§10.5).

Two candidates carry roughly a third of the occurrence mass between them. Counted as whole words over the full text of all 343 reports rather than inside the four audited ones, *compliance* occurs 55,780 times and *social* 48,198. Neither is admissible as a bare term, and they fail differently (Table 9). Of *social*’s occurrences 60.3% sit in unambiguously ESG context and only 8.7% unambiguously outside it, concentrated in three enumerable strings — *social security*, *social media*, *social network* — so on the evidence the right treatment of *social* is an exception list. The vocabulary architecture supplies none: a bare entry in a TF-IDF vocabulary cannot carry a negative lookahead, and no exclusion mechanism exists at any stage. What shipped for *social* is therefore the opposite of what its own analysis recommends — an inclusion list of eight collocations, admitting only the uses they name and discarding the long tail of valid ones the analysis identified as the majority of its mass. *Compliance* inverts the evidence: 41.8% are unambiguously ESG, 8.7% are accounting or audit usage sitting almost entirely in the auditor’s report and the notes to the financial statements, and the largest block, 44.5%, carries no marker either way, so for that term an inclusion list of fifteen collocations is what the evidence supports as well as what was built. *Proxy* was adjudicated on the same evidence, against *proxy hedging*.

These shares are measured over the full text of each report, not over the extracted zones, and the two bases differ in exactly the direction that matters: extraction has already discarded most of the notes and the auditor’s report, which is where the accounting sense of *compliance* lives. The false-positive rate inside the pipeline is therefore lower than Table 9 shows, which bounds the problem from above and does not estimate it.

What the dispersion analysis retires is the sector-bias objection, which is now measured, and not the recall bound, which is unchanged: the adopted vocabulary incorporates none of the 147, and every figure in this paper is computed on the 154 entries. The bound stands exactly where §12 leaves it.

Table 9: Context of the two largest audit candidates, classified by surrounding vocabulary over the full text of all 343 reports. Shares of all occurrences of each term, per cent.

Context	<i>social</i>	<i>compliance</i>
Unambiguously ESG	60.3	41.8
Mixed	10.6	3.8
No marker either way	20.4	44.5
Not ESG: payroll, tax, communications	7.1	—
Not ESG: accounting or audit	1.6	8.7
Not ESG: other	—	1.2
Total	100.0	100.0

6 Index specification

This section defines every quantity the paper computes. It is the reference definition: later sections select among the specifications given here (§9), test sensitivity to that selection (§10) and report the resulting series (§11), but none of them redefines anything.

6.1 Notation and the two text representations

Let $d = 1, \dots, n$ index the $n = 343$ reports of §3 and $j = 1, \dots, V$ the entries of the lemmatised ESG vocabulary of §5, with $V = 154$. Preprocessing emits two representations of each document, and the components are not all computed on the same one:

- *lemmatised text* — lowercased, stopword-filtered, punctuation- and numeral-stripped, lemmatised (§5); whitespace token count ℓ_d , with $L_d = \max(\ell_d, 1)$;
- *raw extracted text* — the ESG zones of §4 with whitespace collapsed and nothing else removed, digits intact; whitespace token count r_d , with $R_d = \max(r_d, 1)$.

QUANT is computed on the raw extracted text; SUS, SEN, HEDGE and BREADTH are computed on the lemmatised text. The split is not incidental. Preprocessing discards numeric tokens, so on lemmatised text the **percentages** and **large_numbers** families of QUANT match nothing and return exactly zero for every document, without error. Every length-normalised component is divided by the token count of the representation it is computed on — R_d for QUANT, L_d for the others — and the two counts are not interchangeable.

Let c_{dj} be the raw count of vocabulary entry j in the lemmatised text of d , obtained with a fixed-vocabulary count vectoriser over token sequences of length 1 to 3; let $df_j = |\{d : c_{dj} > 0\}|$ and

$$\text{idf}_j = \ln \frac{1+n}{1+df_j} + 1, \quad (2)$$

scikit-learn’s smoothed inverse document frequency (Pedregosa et al., 2011). Supplying the vocabulary in advance disables document-frequency and feature-count filtering, so the **max_df**, **min_df** and **max_features** settings the original discloses (§2) are inoperative here: all V entries are retained and scored in every document. Cross-sectional rescaling of a finished component score x is

$$Z(x)_d = \frac{x_d - \bar{x}}{\sigma_x}, \quad \sigma_x = \sqrt{\frac{1}{n} \sum_d (x_d - \bar{x})^2}, \quad (3)$$

with $Z(x) \equiv 0$ when $\sigma_x = 0$. Both $\bar{x}$ and σ_x are taken over the n documents analysed, so every index value below is a position within this corpus, not an absolute quantity.

6.2 The substance component: three specifications

The original does not determine a single construction of SUS: it discloses **TfidfVectorizer** settings but neither the vector normalisation nor the aggregation from per-term weights to a document score (§2). We therefore compute three, differing in term weighting and in vector normalisation:

$$\text{SUS}_d^{\text{dens}} = 100 \cdot \frac{\sum_j c_{dj}}{L_d}, \quad (4)$$

$$\text{SUS}_d^{\text{tfidf-}\ell} = 100 \cdot \frac{\sum_j c_{dj} \text{idf}_j}{L_d}, \quad (5)$$

$$\text{SUS}_d^{\text{tfidf-L2}} = \frac{1}{V} \sum_j \frac{c_{dj} \text{idf}_j}{\|\mathbf{c}_d \odot \mathbf{idf}\|_2}, \quad (6)$$

where $\mathbf{c}_d \odot \mathbf{idf}$ is the elementwise product of the count vector with the idf vector, and $\text{SUS}_d^{\text{tfidf-L2}} = 0$ by convention when a document contains no vocabulary entry. The subscripts name the scaling applied to the term vector: ℓ for division by document length, L_2 for vector normalisation, that is scaling the term vector to unit L_2 norm. Equation (4) is ESG mentions per 100 lemmatised tokens under uniform term weights. Equation (5) applies idf weights to the same length-normalised construction. Equation (6) is our reading of the original: idf weights, L_2 vector normalisation of the document’s term vector, mean over the vocabulary. The factors 100 and $1/V$ are positive affine rescalings and are annihilated by (3); the sum-versus-mean choice in (6) is therefore immaterial, and the specifications differ only in weighting and in vector normalisation. Both idf specifications estimate df_j on the analysed sample and are corpus-dependent; (4) is not.

6.3 Tone, quantification, hedging and breadth

Sentiment. SEN is Loughran–McDonald polarity (Loughran and McDonald, 2011). Let s_d^+ and s_d^- count the tokens of the lemmatised text whose Porter stem belongs to the positive (140 stems) and negative (893 stems) lists respectively:

$$\text{SEN}_d = \frac{s_d^+ - s_d^-}{s_d^+ + s_d^- + \varepsilon}, \quad \varepsilon = 10^{-6}. \quad (7)$$

Here ε only prevents division by zero. This is a bounded ratio on $[-1, 1]$, not a density: up to ε it depends only on the balance of s_d^+ and s_d^- , not on how many sentiment-bearing tokens the document contains, and it is 0 when there are none.

Quantification. QUANT is the density of regular-expression hits from four families (Table 10) over the *raw* extracted text, matched case-insensitively:

$$\text{QUANT}_d = \frac{1}{R_d} \sum_{g \in G} m_g(\text{raw}_d), \quad (8)$$

with m_g the number of matches of family g and $G = \{\text{percentages}, \text{large_numbers}, \text{units}, \text{frameworks}\}$.

Table 10: The four QUANT regular-expression families.

Family	Pattern
percentages	<code>\b\d+(?:[.,]\d+)?\s*%</code>
large_numbers	<code>\b(?:19 20)\d{2}\b\d{4,}\b</code>
units	<code>\b(?:tonne ton mt kto2 co2e? ghg kwh mwh gwh twh mw gw litre liter m3 cubic meter)\b</code>
frameworks	<code>\b(?:gri tcfd sasb issb sdg ungc sfdr csrd un global compact paris agreement taxonomy)\b</code>

The **large_numbers** family excludes four-digit strings of the form 19xx and 20xx, so that citing dates does not register as quantification.

Hedging. HEDGE is the share of lemmatised tokens belonging to the hedge lexicon H of §5 ($|H| = 1,263$ forms, the Uncertainty, WeakModal, StrongModal and Constraining categories of Loughran and McDonald, 2011, lemmatised):

$$\text{HEDGE}_d = \frac{|\{t \in \text{tokens}(\text{lem}_d) : t \in H\}|}{L_d}. \quad (9)$$

Breadth. BREADTH is the effective number of distinct ESG terms in a document: the exponential of the Shannon entropy of the term-share distribution, i.e. a Hill number of order one (Hill, 1973). With $p_{dj} = c_{dj} / \max(\sum_k c_{dk}, 1)$,

$$\text{BREADTH}_d = \exp\left(-\sum_{j: p_{dj} > 0} p_{dj} \ln p_{dj}\right) \in [1, V]. \quad (10)$$

It equals 1 when all mentions fall on one term — and, by the $\max(\cdot, 1)$ convention in p_{dj} , also when a document contains no vocabulary entry at all — and V when mentions are spread uniformly across the vocabulary. **BREADTH is not a component of either index.** It is reported alongside them because it is the quantity $\text{SUS}^{\text{tfidf-}L^2}$ turns out to measure (§7); reporting it as a separate column is what allows that claim to be checked against the published results rather than taken on assertion.

6.4 The composite indices

$$\text{ESGSI}_d = Z(\text{SEN})_d - Z(\text{SUS})_d, \quad (11)$$

$$\text{ESGSI}_d^{\text{ext}} = Z(\text{SEN})_d - Z(\text{SUS})_d - w_q Z(\text{QUANT})_d + w_h Z(\text{HEDGE})_d, \quad (12)$$

with $w_q = w_h = 0.5$. The signs follow the construction: quantified disclosure counts against a washing reading and is subtracted, hedged disclosure counts towards one and is added. The weights are set by fiat and are not calibrated (§12). Equation (11) takes **SUS** from one of (4)–(6); we write $\text{ESGSI}(\text{SUS}^{\text{dens}})$ and so on where the choice matters. The labelling rule is the original’s, and is definitional rather than estimated: $\text{ESGSI}_d > 0$ yields *Potential ESGwashing*, $\text{ESGSI}_d \leq 0$ yields *Likely Genuine* (Lagasio, 2024, pp. 4–5). The unhyphenated spelling is the original’s and is used wherever its rule is quoted; our pipeline writes the same label as *Potential ESG-washing*, which is the string that appears in our output file (§A).

Equation (11) reproduces the formula printed by Lagasio (2024, p. 4),

$$\text{ESGSI}_i = \frac{\text{SEN}_i - \overline{\text{SEN}}}{\sigma_{\text{SEN}}} - \frac{\text{SUS}_i - \overline{\text{SUS}}}{\sigma_{\text{SUS}}},$$

while the prose of the same subsection states that “both sentiment and sustainability scores are normalized to a uniform scale using min-max normalization” (Lagasio, 2024, p. 4). A difference of z-scores and a difference of min–max-rescaled scores are not monotone transforms of one another. We adopt the printed formula throughout; the contradiction is recorded here at the point of definition and adjudicated in §8.

Table 11 collects every constant and every representation choice on which the definitions above depend; the full lexicons, versions and seeds are in §A. That appendix reprints the same values in Table 31, adding the extraction and preprocessing stages, so that a reader has one consolidated listing. The two tables are not independent sources: the values are defined here, and the appendix table is a consolidation of this one together with §4 and §5. Any disagreement between them would be an error in the appendix.

7 Measurement critique: TF-IDF as a representation, not an instrument

TF-IDF was designed for information retrieval (Spärck Jones, 1972; Manning et al., 2008). Retrieval ranks documents against a query, so a document’s absolute scale is a nuisance parameter: what matters is whether this document is a better match than that one, and a report that says the same things twice as often is not a better match. Measurement asks a different question —

Table 11: Parameters of the index specification.

Parameter	Value	Applies to
Text representation	lemmatised raw extracted	SUS, SEN, HEDGE, BREADTH QUANT
Vocabulary size V	154 lemmatised entries	SUS, BREADTH
n -gram range	(1, 3), from the longest entry	SUS, BREADTH
Document-frequency filtering	none; vocabulary fixed in advance	all SUS, BREADTH
Term-frequency transform	raw counts, no sublinear damping	all SUS
idf weights	smoothed, Eq. (2)	$SUS^{\text{tfidf-}\ell}$, $SUS^{\text{tfidf-}L2}$
Vector normalisation	none L2, within document	SUS^{dens} , $SUS^{\text{tfidf-}\ell}$ $SUS^{\text{tfidf-}L2}$
Aggregation over j	sum mean over V	SUS^{dens} , $SUS^{\text{tfidf-}\ell}$ $SUS^{\text{tfidf-}L2}$
Length denominator	L_d (lemmatised tokens) R_d (raw tokens) none	SUS^{dens} , $SUS^{\text{tfidf-}\ell}$, HEDGE QUANT $SUS^{\text{tfidf-}L2}$, SEN, BREADTH
Hedge lexicon size	1,263 forms	HEDGE
Polarity smoothing ε	10^{-6}	SEN
Score normalisation	z-score, population σ	ESGSI, ESGSI ^{ext}
Extended-index weights	$w_q = w_h = 0.5$	ESGSI ^{ext}
Label threshold	ESGSI > 0	both indices

how much of a construct a document contains — and there the scale *is* the quantity of interest. The object of this section is the representation as it comes out of the box: idf term weights, and the unit-length scaling of the document vector that is the library default. Both are calibrated for the first question and neither is safe for the second, but they are not equally unsafe. **The vector normalisation is first-order and the term weighting is second-order**, and the section is ordered accordingly. The argument concerns a class of methods, it holds analytically, and it would hold unchanged if the original had documented every setting it used. It is not a claim about the original’s conclusion; that claim belongs to §8 and rests on evidence of a different kind.

One attribution point governs everything below, and we concede it at the outset rather than leave a referee to raise it. The original (Lagasio, 2024) discloses four `TfidfVectorizer` settings and never mentions `norm` (§2). The four it discloses are exactly four non-default arguments, so a reader following the ordinary convention that only departures from library defaults are reported would infer that the remaining defaults — `norm='l2'` among them — were in force in the library version the article names (Pedregosa et al., 2011). We make that inference and we hold it to be reasonable. Having relied on it here, we do not also count the silence on `norm` as a reproducibility failure: §2.4 rests that charge on the aggregation rule, which is a modelling step rather than a library setting and which no default convention can supply, on the unpublished keyword list, and on the contradiction between the original’s prose and its printed formula on score normalisation. Those three are unanswerable from any convention; this one is not, and we say so in both places.

The inference is nonetheless an *inference*, it is labelled as one wherever it is used, and it can be denied at no cost to the argument: what follows is a property of $SUS^{\text{tfidf-}L2}$ — our reconstruction under that inference (§6) — and of any pipeline built the same way. It is not a documented choice of the original, and we do not assert that the original made it.

7.1 Vector normalisation discards magnitude

Vector normalisation — L2 scaling of a document’s term vector to unit length, applied within the document at the TF-IDF stage — retains the vector’s direction, that is the *mix* across terms, and discards its length, that is *how much* there is. The algebra is immediate: for any $\mathbf{v} \neq \mathbf{0}$ and any scalar $c > 0$,

$$\frac{c\mathbf{v}}{\|c\mathbf{v}\|_2} = \frac{c\mathbf{v}}{c\|\mathbf{v}\|_2} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}. \quad (13)$$

Two documents whose ESG term counts are proportional to one another therefore receive *identical* unit vectors. Nothing downstream recovers the difference: the aggregation from per-term values to a document score is then applied to identical inputs and returns identical outputs, whatever that aggregation is — sum, mean, weighted mean or anything else. This is not an approximation or a small-sample effect; it is an identity.

The distinction matters because it separates the two silences in the original’s specification, which are unequal in both respects that matter. They are unequal in what a convention can close: **norm** has a library default and the aggregation rule has none (§2). They are also unequal in consequence, and in the opposite direction. Sum and mean differ by the constant factor $1/V$, and $Z(\cdot)$ annihilates a positive affine rescaling of a component, so on our index the aggregation choice cannot move a single value (§6) — the silence no convention closes turns out to be the harmless one here. The vector normalisation is not of that kind. It does not rescale the score; it changes what the score is a function of, which is why an inference, however reasonable, is a thin thing to leave a first-order choice resting on.

Whether that is a defect depends on the construct being measured. Discarding magnitude is appropriate when the question is which document best matches a query. It is not appropriate here: ESGSI contrasts the tone of a report against the amount of substance behind it, so a substance component that returns the same value for a document and for the same document carrying twenty times the ESG content is measuring on the wrong side of that comparison.

7.2 Two demonstrations

Equation (13) predicts two failures. Both are executed end to end on the live pipeline rather than argued, so that the pipeline is shown to behave as the algebra says rather than assumed to.

Amplification. We take a document synthesised from the observed vocabulary counts of one report — synthesised so that the counts can be scaled exactly without perturbing the n -gram structure — multiply those counts by $k \in \{1, 2, 5, 20\}$, re-embed the result in the remaining reports and re-run the full vectorisation. Scaling a document’s own counts leaves every df_j unchanged, since a term present remains present and a term absent remains absent, so the idf vector is fixed across the four runs and the comparison is clean. Table 12 reports the outcome. Twenty times the ESG content leaves $\text{SUS}^{\text{tfidf-L2}}$ unmoved to machine precision, while any specification with a length denominator rises by the factor kL_1/L_k , as it should.

Table 12: The same document with ESG term counts multiplied by k . $\text{SUS}^{\text{tfidf-L2}}$ is reported relative to its value at $k = 1$: Equation (13) does not depend on the idf vector, so the ratio is exact for any corpus, whereas the level is corpus-dependent. The maximum absolute deviation across the four runs is 6.9×10^{-18} .

k	Tokens	ESG hits	SUS^{dens}	$\text{SUS}^{\text{tfidf-L2}}$, relative to $k = 1$
1	16,556	700	4.2281	1.000000
2	18,112	1,400	7.7297	1.000000
5	22,780	3,500	15.3644	1.000000
20	46,120	14,000	30.3556	1.000000

Dilution. The converse is the case a reader will recognise from practice. We append 56,000 non-ESG filler tokens, drawn from the document’s own frequent non-vocabulary words, to a 22,576-token report, leaving its 695 ESG mentions untouched. Table 13 reports the outcome: measured ESG density falls from 3.0785 to 0.8845 per 100 tokens, a 3.48-fold dilution, and $\text{SUS}^{\text{tfidf-L2}}$ does not move to ten decimal places. A report can be padded without limit — with boilerplate, with repetition, with any material outside the ESG vocabulary — at no cost to its measured sustainability substance.

Table 13: Dilution of a report with non-ESG filler text.

Variant	Tokens	ESG hits	SUS^{dens}	$\text{SUS}^{\text{tfidf-L2}}$
Original	22,576	695	3.0785	0.0347763839
+ 56,000 filler	78,576	695	0.8845	0.0347763839

7.3 What survives vector normalisation is a spread measure

If magnitude is discarded, something else is being measured, and it is worth naming. Let $\mathbf{u} \geq \mathbf{0}$ be a unit vector in $\mathbb{R}^V$, with $V = 154$ the vocabulary size (§6). Then

$$1 \leq \sum_{j=1}^V u_j \leq \sqrt{V}. \quad (14)$$

In Equation (14) the lower bound follows from $(\sum_j u_j)^2 = \sum_j u_j^2 + \sum_{j \neq l} u_j u_l \geq \|\mathbf{u}\|_2^2 = 1$ for non-negative $\mathbf{u}$, with equality when all mass sits on a single term; the upper bound is Cauchy–Schwarz, with equality when the mass is spread evenly over all V terms. Between the two, mass spread evenly over m terms gives $\sum_j u_j = \sqrt{m}$. Aggregating an L2-normalised vector over the vocabulary — $\text{SUS}^{\text{tfidf-L2}}$ is that sum divided by V — therefore rewards *dispersion across terms*, not *amount of content*: a report mentioning many distinct ESG topics once each scores above a report treating a few of them intensively, and the ceiling is set by the size of the vocabulary rather than by the disclosure.

That is what the data show. On our corpus $\text{SUS}^{\text{tfidf-L2}}$ correlates 0.9733 with BREADTH, the effective number of distinct vocabulary terms (§6), and 0.4177 with ESG density. It is, to a correlation of 0.97, a measure of topical breadth carrying the name of a measure of disclosure intensity. Breadth is a legitimate construct and may well be worth reporting; it is simply not the construct the index is defined to capture.

7.4 The term weighting is second-order

Inverse document frequency down-weights terms that appear everywhere, because in retrieval a term present in every document separates nothing. In a corpus of ESG disclosure the same terms are present everywhere for the opposite reason: they are what the genre is about. Rarity and topical centrality are inversely related here, so idf transfers weight away from the vocabulary that carries the disclosure and towards the vocabulary that does not. Table 14 gives the inversion on our corpus.

Table 14: Concentration of ESG term occurrences by idf band ($n = 343$).

Band	Terms	Share of all ESG occurrences
idf < 1.2 (document frequency > 280)	43	78.35 %
idf > 2	46	2.80 %

Forty-three terms hold 78.35 % of all ESG occurrences in the corpus and are weighted at close to the floor of 1; forty-six hold 2.80 % and are weighted at up to nearly five times it. The exchange rate is explicit at the level of individual terms. *Fair condition* occurs eight times in the entire corpus and carries $\text{idf} = 4.8947$; *climate*, among the most frequent entries in the vocabulary, carries $\text{idf} = 1.0088$. One mention of the former is therefore worth 4.852 mentions of the latter. No account of what a sustainability report discloses supports that exchange rate, and no reader of such a report would recognise it.

It is nonetheless the weaker of the two criticisms, and we say so rather than present the pair as equals. The inversion is theoretically indefensible and empirically close to inert, for the reason the table itself supplies: the terms idf promotes carry 2.80 % of the mass. Holding the treatment of length fixed — SUS^{dens} and $\text{SUS}^{\text{tfidf-}\ell}$ both divide by document length, and neither applies vector normalisation (§6) — and varying only the term weighting, the two specifications correlate 0.9926, and their assembled indices disagree about the label of 6 of 343 reports. Varying the vector normalisation instead moves 96 of 343 (§10). A reader who accepts the first-order argument and rejects this subsection in its entirety loses nothing that matters here.

The concession runs further, and stating it is what keeps the first-order claim credible. Nothing above indicts idf weighting as such for measurement use: with the vector normalisation replaced by a length denominator, the same idf weights produce a specification that agrees with our independent yardstick at least as well as any other we compute (§9). What this section establishes is a defect of one library default, not of the weighting scheme and not of the family.

7.5 The sentiment component is exposed to the same class of problem

The critique above concerns one half of the index. The other half is SEN_i , TextBlob polarity, which enters ESGSI with the same weight as the substance component (§6). TextBlob’s polarity is a pattern-matching lexicon whose scores are modified by negation and by intensifiers. The original’s preprocessing, disclosed in its Section 3.1.1 (p. 3), lowercases, strips URLs, numerals and punctuation, removes NLTK’s English stopwords augmented by an unpublished custom list, discards every token shorter than three characters, and lemmatises (Lagasio, 2024). That pipeline removes the tokens those rules operate on — *no* falls below the three-character floor, *not* and *very* are NLTK stopwords — and stopword removal breaks the adjacency the modifier rules read. A polarity score is still returned; it is no longer the quantity the instrument was validated to produce. Whether the raw text or this preprocessed output reached TextBlob is never stated (§2), and the five-fold cross-validation announced for the sentiment model is never reported. The instrument is therefore either applied outside its validity conditions or it is not, and the article does not say which. Our own substitution, a count-based finance dictionary that presupposes no negation or intensifier parsing (§6), is a divergence by design and not a repair.

7.6 What carries into the next section

Both criticisms share a structure. There is a vector-normalisation choice at the TF-IDF stage and a score-normalisation choice at index assembly; at each stage that choice dominates the weighting choice; and the original states neither (§2). The two are not undisclosed in the same way, and the difference cuts against us in the first case and for us in the second: the first is recoverable from a reporting convention, as conceded above, whereas the second is not merely unstated but stated twice and inconsistently, in prose and in a printed formula that specify different estimators. The second stage is the subject of §8. What the present section supplies is the reason to expect a result there: once first-order choices of this kind are left to a convention or to a contradiction, a published conclusion that turns on one of them is a predictable outcome rather than a coincidence.

8 The score-normalisation artefact

The original’s headline empirical conclusion is that ESG-washing is largely absent from the reports it studies: “*The ESGSI predominantly shows negative values, suggesting a general absence of greenwashing practices among the entities studied*” (Lagasio, 2024, p. 6). This section argues that the conclusion is an artefact of two methodological choices, neither of which the paper states in a form a reader could act on — the *score normalisation* applied cross-sectionally to the two components at index assembly, which the paper specifies twice and inconsistently, and the specification of the substance component (§6), whose aggregation rule it never gives at all — rather than a property of the disclosure.

The argument has three parts, and they are deliberately kept separate because they do not carry equal weight. The first establishes, from the original’s own published tables, *which* score normalisation produced its numbers; this rests on arithmetic that requires nothing of our pipeline. The second shows analytically why that choice determines the sign of the reported mean; it too requires nothing of our pipeline, and nothing of the original beyond the two maps themselves. The third demonstrates the consequence on our corpus. The first two are jointly sufficient for the section’s claim and neither uses our data; the third is corroboration, and we mark it as such.

8.1 Which score normalisation produced the published numbers

Section 3.3 of the original states its score normalisation in prose:

“To ensure comparability, both sentiment and sustainability scores are normalized to a uniform scale using min-max normalization. This normalization technique is widely used in text analysis and machine learning to address potential biases due to variations in document length and linguistic style (Sebastiani, 2002).” (Lagasio, 2024, p. 4)

The formula printed a few lines below, in the same subsection, is a difference of z -scores:

$$\text{ESGSI}_i = \frac{\text{SEN}_i - \overline{\text{SEN}}}{\sigma_{\text{SEN}}} - \frac{\text{SUS}_i - \overline{\text{SUS}}}{\sigma_{\text{SUS}}} \quad (15)$$

with the six symbols defined immediately beneath it as the component scores, their sample means and their sample standard deviations (Lagasio, 2024, p. 4). The paper never reconciles the two; we flagged the contradiction at the point of definition (§6) and resolve it here.

Four lines of evidence, all drawn from the original’s own Table 2 ($n = 749$), settle it against the printed formula.

Support. The table notes state verbatim that the “*Normalized Sustainability Score: Scores range from 0 (low sustainability) to 1 (high sustainability)*” and that the “*Normalized Sentiment Score . . . [is] normalized between 0 and 1*” (Lagasio, 2024, p. 6); the note to Figure 1 repeats that the plotted component scores run “*over a continuous range from 0 to 1*” (Lagasio, 2024, p. 5). The reported minima, 0.0546 for the sustainability component and 0.0000 for the sentiment component, are both non-negative. A z -scored variable has mean zero and positive dispersion, so its minimum is necessarily negative. Neither component can have been z -scored.

Dispersion. Table 2 gives standard deviations of 0.1321 for the sustainability component and 0.0853 for the sentiment component. A z -score has unit variance by construction. Two variables with standard deviations of 0.1321 and 0.0853 have not been z -scored.

Location. Table 2 reports mean $\text{ESGSI} = -0.2179$ with standard deviation 0.1614. The sentiment component has mean 0.3727 and the sustainability component mean 0.5906, and

$$0.3727 - 0.5906 = -0.2179, \quad (16)$$

which is the reported mean exactly. Under Equation (15) the sample mean of ESGSI is 0 by construction, since it is a difference of two mean-zero variables. It is not 0.

Quantiles. Table 2 also reports quartiles. For the index they are -0.3245 , -0.2177 and -0.1054 (Lagasio, 2024, p. 6): the third quartile lies below the labelling threshold, so at least three quarters of the 749 reports fall at or below -0.1054 , and the share the original’s own instrument would have labelled “*Potential ESGwashing*” is bounded above at roughly one report in four — at most 188 of 749 on the most conservative of the standard quantile conventions. The original never reports that share (§2); its own table bounds it. The bound is hard to reconcile with Equation (15). Under the printed formula the index has mean zero by construction, so for only a quarter of the sample to exceed zero the index would have to be pronouncedly right-skewed, with its mean at or above its own third quartile. Nothing in Table 2 supplies such a skew: the reported quartiles are close to evenly spaced about the median for the sustainability component (0.5020/0.5876/0.6732, gaps of 0.0856 and 0.0856), for the sentiment component (0.3188/0.3668/0.4169, gaps of 0.0480 and 0.0501), and for the reported index itself, whose mean (-0.2179) and median (-0.2177) differ by 0.0002.

The identity in Equation (16) is not confined to the overall row. Subtracting the reported “Normalized Sustainability Score” from the reported “Normalized Sentiment Score” reproduces the reported ESGSI in every one of the twenty-eight rows the original publishes across Tables 2–6: exactly in twenty-four of them, and to within one unit in the fourth decimal place — the resolution at which the components themselves are printed — in the remaining four (Table 15). No row was selected or omitted; the table below is the complete published set.

One passage in the original describes its index in terms that fit the printed formula rather than the tables. Introducing Figure 1, the paper states that the distribution “*is bell-shaped and centered around an ESGSI of zero*” and that “*a symmetric dispersion pattern around the central value indicates an equivalent propensity for documents to exhibit both potential greenwashing behaviors and genuine sustainability representation*” (Lagasio, 2024, p. 6). That description is not consistent with Table 2 on the same page — mean -0.2179 , median -0.2177 , third quartile -0.1054 , all below zero — nor with the paper’s own summary of that table, two paragraphs later, that the index “*predominantly shows negative values*”. We do not attempt to explain the discrepancy, and we do not treat prose as evidence about which map was applied: the determination above rests on the numeric tables, which are also what every downstream analysis in the original — Tables 3–7 and the ANOVA — is computed from.

The estimated index is therefore the difference of the two reported components on their $[0, 1]$ scale, the min–max rescaling of the original’s own prose, and not the z -score difference the paper prints. What the four arguments establish jointly is the negative, and it is the negative that the section needs: the published numbers were not produced by Equation (15). Nothing in that conclusion requires the rescaling to have been min–max in every operational detail. It requires only that the components be bounded and that the index be their plain difference, and the original’s tables show both directly.

This argument uses only numbers the original published. It would stand unchanged if our corpus, our vocabulary and our pipeline did not exist, and it is not revisited, reinforced or qualified anywhere later in this paper. The choice is established here, and only here.

8.2 Why the choice determines the sign of the result

Write $Z(\cdot)$ for the z -score map and $M(\cdot)$ for the min–max map applied cross-sectionally to a finished component score. Both are affine and increasing, so within a component they are rank-equivalent; they are not equivalent once two components are differenced, because they equalise the two components’ dispersions in different ways.

Under Z , the sample mean of $Z(\text{SEN}) - Z(\text{SUS})$ is 0 identically, whatever the two components contain: the estimator centres the index on its own labelling threshold before any document is read. What remains free is only the shape of the distribution about that centre, so whether a majority falls below zero becomes a question about the skew of a mean-zero variable rather than about disclosure practice — and the original’s own reported quartiles (§8.1) describe compo-

Table 15: Reconstruction of the original’s reported ESGSI as the difference of its two reported components. Every row published in Lagasio (2024), Tables 2–6, is included. Four rows — Consumer Staples, the Table 4 total, and the second and fourth ESG quartiles — differ by one unit in the fourth decimal place, which is the resolution at which the two components are printed. All figures are read from the original.

Row	Norm. SEN	Norm. SUS	Difference	Reported ESGSI
<i>Table 2, p. 6 — full sample</i>				
Overall	0.3727	0.5906	−0.2179	−0.2179
<i>Table 3, p. 7 — by region</i>				
Europe	0.3477	0.6144	−0.2667	−0.2667
North America	0.3917	0.5731	−0.1814	−0.1814
South America	0.3229	0.6622	−0.3393	−0.3393
Other	0.3665	0.5760	−0.2095	−0.2095
Total	0.3572	0.6064	−0.2492	−0.2492
<i>Table 4, p. 7 — by GICS sector</i>				
Communication Services	0.3637	0.5909	−0.2272	−0.2272
Consumer Discretionary	0.4087	0.5889	−0.1802	−0.1802
Consumer Staples	0.4048	0.6138	−0.2090	−0.2089
Energy	0.3478	0.6595	−0.3117	−0.3117
Financials	0.3460	0.5944	−0.2484	−0.2484
Health Care	0.3832	0.5775	−0.1943	−0.1943
Industrials	0.3850	0.5743	−0.1893	−0.1893
Information Technology	0.3788	0.5547	−0.1759	−0.1759
Materials	0.3584	0.6187	−0.2603	−0.2603
Real Estate	0.3634	0.5672	−0.2038	−0.2038
Utilities	0.3483	0.5873	−0.2390	−0.2390
Total	0.3716	0.5934	−0.2218	−0.2217
<i>Table 5, p. 8 — by total-assets quartile</i>				
1st quartile	0.3853	0.5688	−0.1835	−0.1835
2nd quartile	0.3829	0.5878	−0.2049	−0.2049
3rd quartile	0.3620	0.5949	−0.2329	−0.2329
4th quartile	0.3595	0.6093	−0.2498	−0.2498
Total	0.3724	0.5902	−0.2178	−0.2178
<i>Table 6, p. 9 — by ESG-score quartile</i>				
1st quartile	0.3857	0.5650	−0.1793	−0.1793
2nd quartile	0.3772	0.5898	−0.2126	−0.2127
3rd quartile	0.3751	0.5855	−0.2104	−0.2104
4th quartile	0.3529	0.6223	−0.2694	−0.2695
Total	0.3727	0.5907	−0.2180	−0.2180

nents that supply no such skew. Under the printed formula, “predominantly negative” is neither available as a result nor interpretable as one. Under M , the sample mean is

$$\overline{M(\text{SEN})} - \overline{M(\text{SUS})}, \quad (17)$$

which the construction pins to no particular value; it is fixed by where each component’s mass sits inside its own observed range. Equation (17) is a statement about the position of two marginal distributions within their ranges, not about firm behaviour. The original reports the two quantities directly — substance 0.5906, above the midpoint of its own reported range, against sentiment 0.3727, well below the midpoint of its — and their difference is the reported index mean, by Equation (16). The original describes the same asymmetry in words, calling its sustainability scores “*skewed towards higher values*” and its sentiment scores approximately bell-shaped (Lagasio, 2024, p. 6). The sign of the mean is therefore a fact about where two marginal distributions sit within their own ranges. It becomes a claim about firm behaviour only through the zero threshold, which is definitional and uncalibrated (§6) and which inherits the asymmetry directly.

The paper’s own justification for the choice does not survive inspection. The sole stated defence is that min–max score normalisation addresses “*potential biases due to variations in document length and linguistic style*” (Lagasio, 2024, p. 4), a claim repeated at the end of its Section 3.2.3. Min–max score normalisation is a single monotone affine rescaling applied across the cross-section, $(x - \min x)/(\max x - \min x)$. It is rank-preserving within each variable, so whatever dependence on document length is present in SUS_i before the rescaling is present, in the same rank order, after it. A rank-preserving cross-sectional map cannot remove a dependence that survives in the ranks. The operation that could — vector normalisation, applied to the document’s term vector at the TF-IDF stage, whose length-neutrality we demonstrate in §7 — is credited with nothing and is never named (§2). The paper’s only stated protection against length bias is attributed to the operation that cannot provide it, while the operation that can provide it goes unmentioned.

8.3 The consequence, demonstrated on our corpus

We can now ask what the two choices do jointly. Because our corpus, ESG vocabulary and sentiment dictionary all differ from the original’s by design (§1, §5), what follows is a demonstration of a mechanism on an independent sample; it is not a replication of the original’s estimate and is not offered as further evidence for §8.1.

Table 16 crosses the two score normalisations with two substance specifications — our density specification SUS^{dens} and the L2 vector-normalised TF-IDF specification $\text{SUS}^{\text{tfidf-L2}}$ that follows the original’s weighting (§6) — over our 343 reports.

Table 16: Mean index and number of reports flagged ($\text{ESGSI} > 0$) under the 2×2 factorial over score normalisation and substance specification, $n = 343$. The two z -score means are zero by construction and are printed at the same resolution as the others. Counts are quantities of our corpus and are not estimates of any quantity of the original’s sample.

Score normalisation	Substance specification	Mean	Flagged
z -score	SUS^{dens}	0.0000	171/343
z -score	$\text{SUS}^{\text{tfidf-L2}}$	0.0000	159/343
min–max	SUS^{dens}	−0.0004	166/343
min–max	$\text{SUS}^{\text{tfidf-L2}}$	−0.2388	54/343

Only the joint configuration moves the mean appreciably away from zero; the other three cells sit within 0.0004 of it. Holding the substance specification at $\text{SUS}^{\text{tfidf-L2}}$ and switching the

score normalisation to z -scores returns the mean to zero exactly; holding min-max and switching the substance specification to SUS^{dens} returns it to within 0.0004 of zero. The flagged count falls from about half the corpus to 54/343 only in the joint cell.

Three restrictions on how this may be read. First, our -0.2388 and the original’s -0.2179 differ by 0.0209, and the correspondence between them is consistency, not proof: a matching sign and order of magnitude obtained on a different corpus with a different vocabulary corroborates the mechanism and does not establish it, and coincidence is possible. The claim we make is that these two choices jointly reproduce the statistic and are jointly necessary for it *on our corpus* — not that we have shown what the original ran, which §8.1 establishes on stronger and independent grounds. Second, the flagged counts of 171/343 and 54/343 are quantities of our corpus alone and are not estimates of the original’s flagged share. The original never reports that share (§2) and nothing in this subsection recovers it; the only statement available about it is the upper bound its own published quartiles impose, and that bound was obtained in §8.1 without reference to anything here. Third, a reader who rejects this subsection in its entirety is left with §8.1 and §8.2 intact, and those are sufficient for the section’s claim.

8.4 What follows

The published conclusion does not survive the paper’s own printed formula. Under Equation (15) the index is centred on its own labelling threshold by construction, so “predominantly negative” is not a substantive result the estimator can return; under the min-max rescaling actually used, the sign of the mean is set by the position of two marginal distributions within their own ranges and by the substance specification — the first specified twice and inconsistently, the second not specified at all. The conclusion is thus a property of the index as assembled, and it is not stable under alternatives that the paper’s own text licenses — which is the empirical counterpart of the label instability we report in §10.

This is a claim about the instrument, and its scope should not be overread. We do not claim that ESG-washing is prevalent, nor that the firms in the original’s sample engage in it; establishing either would require an external benchmark that neither the original nor this paper possesses (§12). What we claim is narrower and, we think, harder to avoid: the reported absence of ESG-washing is a consequence of the score normalisation applied and of how substance was specified, and neither choice was disclosed in a form that would have let a reader see it.

9 Convergent validity and specification choice

§2 established that Lagasio (2024) states neither how per-term TF-IDF values are aggregated into a document-level substance score nor whether the term vector is normalised, and §7 that the second of those silences is first-order: it determines what the resulting quantity measures. Three substance specifications therefore remain live and no reading of the original’s text selects among them. This section selects on evidence, using a criterion external to the specifications compared — external, but not independent of them, and the two respects in which it is not are stated below with their magnitudes rather than deferred to §12. One of the two is then removed by recomputation on bases the two instruments do not share (§9.3); the other is not removable within this design and is reported as a bound.

9.1 QUANT as a yardstick, and the senses in which it is not independent

QUANT (§6) is a density of regular-expression matches over the *raw* extracted text: percentage expressions, integers of four or more digits with calendar years excluded, physical units of mass, energy and volume, and reporting-framework acronyms, divided by the raw token count R_d . Its construction shares almost nothing with any SUS specification. There is no vectoriser, no lemmatised vocabulary, no document-frequency weighting and no vector space; the token base is

the raw text with digits intact, whereas every SUS specification is computed on the lemmatised, stopword-filtered text (§5).

It also targets a property that substantive disclosure should exhibit: a report that discloses substance quantifies it, in tonnes, percentages and absolute counts, and names the frameworks under which those figures are computed. Agreement with QUANT is therefore a convergent-validity test in the sense of Campbell and Fiske (1959) — two instruments built by largely unrelated means, aimed at overlapping constructs, should covary. Agreement does not prove validity, since QUANT is itself a crude instrument and two measures can covary for irrelevant reasons; but a substance measure that does not covary with independently counted quantitative content is hard to defend.

“Almost nothing” is doing work in that claim, and we discharge it before the result rather than after it. Independence of construction fails in one respect — fourteen vocabulary entries are counted by both instruments — and independence of sampling in another: both are computed over the same extracted text, with denominators that are two measures of the same document’s length. Both are quantified below, and neither is left to §12.

Table 17: Correlation of each substance specification with QUANT ($n = 343$).

Substance specification	Pearson r	Spearman ρ
SUS ^{tfidf-ℓ}	0.6817	0.7220
SUS ^{dens}	0.6663	0.7076
SUS ^{tfidf-L2}	0.3217	0.4299
Sublinear TF, length-normalised	+0.0067	—
Sublinear TF, mean over vocabulary	+0.5262	—

The two length-normalised specifications track independently measured quantitative content roughly twice as well as the L2-normalised specification does. The ordering is unchanged under rank correlation, so it is not produced by a handful of extreme documents, though the gap narrows there (0.7076 against 0.4299). This is the predicted consequence of the invariance identity in §7: SUS^{tfidf-L2} discards the magnitude of a document’s ESG content and retains only its direction across the vocabulary, and quantified content is a magnitude.

9.2 The yardstick is partly contaminated

The specification choice in this paper rests on agreement with QUANT, and QUANT is not clean: it counts some of the same strings that the SUS vocabulary counts. The instrument doing the validating and the instrument being validated share terms. We give the size of the overlap, the reason the obvious defence of it fails, and then, in §9.3, the computation that settles what the defence cannot.

The **frameworks** group of QUANT matches 11 entries of the 154-term ESG vocabulary (*csrd*, *gri*, *issb*, *sasb*, *sdg*, *sfd*, *tcfd*, and four members of the *taxonomy* family); the **units** group matches three more (*co2*, *co2e*, *ghg*). Those 14 shared entries account for 67,194 of 895,996 vocabulary occurrences over the deduplicated corpus, i.e. **7.50 %** of total SUS occurrence mass. The exposure on the QUANT side we can only bound: the **frameworks** group, which holds 11 of the 14 shared entries but also patterns with no vocabulary counterpart (*un global compact*, *paris agreement*), supplies 17.22 % of all QUANT hits pooled over the corpus, and the **units** group adds the remaining three. We do not report the shared entries’ exact share of QUANT hits, and it should not be read off the group totals. What the group totals do show is that the overlap is not stationary: the **frameworks** share rises from 10.15 % of QUANT hits in 2018 to 21.87 % in 2024, so the shared component grows across the study window and must be carried into §11.

There is a tempting defence, and it does not hold. It runs: the 14 entries belong to the vocabulary, every specification is computed over that same vocabulary, so the overlap inflates

every row of Table 17 alike and cannot generate the ordering between rows. Two things are wrong with it. First, 7.50 % is an *occurrence-mass* share, which is the weight those entries carry under uniform weighting — that is, under SUS^{dens} and under no other specification in the table. IDF weighting penalises terms present in nearly every document, and framework acronyms, *co2* and *ghg* are exactly such terms, so the shared entries carry less than their mass share under $\text{SUS}^{\text{tfidf-}\ell}$ and $\text{SUS}^{\text{tfidf-L2}}$; and under L2 scaling a term present in almost every document contributes little cross-sectional variance to a score that has been stripped of magnitude. Second, and more basic: what displaces a correlation is the covariance the shared terms carry *within each specification*, and a share of occurrences bounds no covariance. No arithmetic converts 7.50 % of mentions into a maximum displacement of r , so the claim that a shared component of that size could not manufacture the distance between 0.6663 and 0.3217 would be an assertion and not a calculation.

The plausible direction is against us. Both mechanisms above reduce the shared entries’ leverage under $\text{SUS}^{\text{tfidf-L2}}$ relative to SUS^{dens} , so if the overlap displaces the table at all it plausibly lifts the top rows further than the bottom row — inflating the very gap this section reports. Every correlation in Table 17 is therefore an upper bound on the level of convergent validity, and we do not defend the ranking by argument.

9.3 The table recomputed on disjoint bases

One computation settles what argument cannot, and we ran it. We deleted the 14 shared entries from the ESG vocabulary, leaving 140 terms, and restricted QUANT to its two purely numeric families, **percentages** and **large_numbers**, dropping **frameworks** and **units** in their entirety. The two instruments then share no string. The restricted QUANT correlates 0.8738 with the full one, so it is a substantially different criterion and not a cosmetic variant of the same one. Table 18 recomputes the comparison on those disjoint bases.

Table 18: Correlation with QUANT on the contaminated and on the fully disjoint bases ($n = 343$). Disjoint bases: ESG vocabulary reduced from 154 to 140 entries, QUANT restricted to **percentages** and **large_numbers**. Pearson r throughout.

Substance specification	Full bases	Disjoint bases
$\text{SUS}^{\text{tfidf-}\ell}$	0.6817	0.4125
SUS^{dens}	0.6663	0.4112
$\text{SUS}^{\text{tfidf-L2}}$	0.3217	0.1799
Ratio, SUS^{dens} to $\text{SUS}^{\text{tfidf-L2}}$	$2.07\times$	$2.29\times$

Read it precisely, because it concedes one thing and establishes another. Every absolute correlation falls, by roughly two fifths — so the overlap does inflate the level of agreement, the figures in Table 17 are upper bounds exactly as the previous subsection said, and the level of convergent validity on clean bases is moderate rather than strong. What the overlap does *not* do is manufacture the ordering. The two length-normalised specifications remain roughly twice as close to independently counted quantitative content as the L2-normalised one, and the relative distance is slightly *larger* once the shared strings are gone, $2.29\times$ against $2.07\times$. That is the opposite of what the previous subsection expected: on the mechanisms set out there the overlap should have widened the gap, and in fact it was compressing it slightly. We record that we had the sign wrong, which is the reason for running the computation rather than arguing about it. The ranking on which this section’s selection rests survives the removal of everything the two instruments had in common; its level does not, and we do not claim it.

The narrow gap between $\text{SUS}^{\text{tfidf-}\ell}$ and SUS^{dens} narrows further, to 0.0013, which reinforces rather than disturbs the reasoning of §9.7: the criterion separates the length-normalised pair from $\text{SUS}^{\text{tfidf-L2}}$ and does not separate the pair.

Neither table, in any case, reaches the part of the diagnosis that uses no yardstick at all. The invariance identity of §7 is algebraic rather than correlational: $\text{SUS}^{\text{tfidf-}L2}$ returns the same value for any two documents whose ESG counts are proportional — demonstrated end to end at a deviation of 6.9×10^{-18} under twentyfold amplification, and unchanged to ten decimal places under a 3.48-fold dilution that drops measured density from 3.0785 to 0.8845 per 100 words. And $\text{SUS}^{\text{tfidf-}L2}$ correlates 0.9733 with BREADTH, a diversity statistic computed with no regex, no raw text and no QUANT. Convergent validity here corroborates a diagnosis that was reached without it. The levels in Table 17 are inflated; the ordering is not, and the diagnosis does not stand on either table alone.

9.4 Independence of construction is not independence of sampling

QUANT and SUS are computed over the same extracted zones (§4). Neither instrument sees text the other does not. A report from which extraction recovered little usable text scores low on both for reasons that have nothing to do with disclosure quality, and any bias in the extraction stage — its density threshold, its context window — propagates identically to both sides of the comparison. The test establishes convergent validity conditional on the extraction stage, not agreement with anything outside our pipeline. We have no external ESG-washing benchmark; neither does the original (§12).

The denominators sharpen the same point. QUANT is a count over raw tokens R_d ; SUS^{dens} and $\text{SUS}^{\text{tfidf-}\ell}$ are counts over lemmatised tokens L_d . These are not the same variable, but they are two measures of the length of one document and they vary together, and two ratios with collinear denominators correlate for that reason alone. The specifications that win the test are the ones built like the yardstick, and $\text{SUS}^{\text{tfidf-}L2}$, being scale-free, cannot benefit from that mechanism even in principle.

The corpus bounds how much of the result this can explain. The fourth row of Table 17 is a specification whose cross-sectional variation is dominated by its length denominator — a numerator confined to a narrow range over a length that is not, as the next subsection establishes — and it correlates +0.0067 with QUANT. Were QUANT’s variation across documents driven by its own length denominator, that specification would correlate strongly with it. It does not, which puts the common-denominator artefact near zero on this corpus. That is an argument from the data, not a formal partialling-out of length, which we do not report.

9.5 Why sublinear TF fails, and where

Raw density invites an obvious objection: a report that repeats one commitment ten times is scored as though it disclosed ten times as much. The standard remedy is sublinear term-frequency scaling (Manning et al., 2008), which replaces the count c_{dj} by $1 + \log c_{dj}$ so that the eleventh mention contributes far less than the second. Applied with the same length denominator,

$$\text{SUS}_d^{\text{sub}} = \frac{100}{L_d} \sum_{j: c_{dj} > 0} (1 + \log c_{dj}), \quad (18)$$

the result is not merely weaker but null: $r = +0.0067$ with QUANT, indistinguishable from zero and two orders of magnitude below SUS^{dens} .

The failure has to be localised before anything is concluded from it, because two distinct quantities are involved and they behave in opposite directions. The vocabulary is curated at 154 terms and a document contains on average 87.75 of them, so the sum in the numerator of (18) ranges over a bounded and comparatively narrow interval: log damping compresses each term’s contribution towards its mere presence, and the sum accordingly tracks the number of *distinct* vocabulary terms in the document rather than the volume of mentions. Document length is under no such bound. The sign change is the diagnostic. Taken before the division, the numerator of (18) moves with the distinct-term count; taken after it, the finished score correlates

−0.3214 with that same count, because variation in L_d now dominates the ratio. The two figures describe different quantities and neither is interpretable without saying which. What a user of the specification would compute is the second: a measure that falls with document length irrespective of content.

The defect is therefore the interaction and not the damping. Aggregated as a mean over the vocabulary instead of divided by document length, sublinear TF correlates +0.5262 with QUANT — the same damping, a different denominator, most of the agreement restored. Sublinear weighting is unusable in this configuration; it is not unusable. We report the distinction because the null result, stated at the level of a method family, would be false.

One kind of evidence is excluded here by design. The specifications also differ in the shape of the temporal series they produce, and that behaviour is used neither for nor against any of them in this section. A specification selected because it delivered the trend the paper later reports would make that trend an artefact of the selection. §11 reports the series and §10 the sensitivity of the index to the specification; the selection uses Table 17 and §7 only.

9.6 The repetition objection, and what would answer it

The objection that motivated sublinear TF is not thereby answered — it is misdirected. Damping treats repetition as measurement noise to be suppressed inside the substance score. But a report that restates the same commitment across five sections is exhibiting a behaviour of interest, plausibly correlated with the construct the index is meant to detect, and folding it into the substance measure discards that information. The appropriate response is to measure redundancy as a quantity in its own right — through the concentration of the term distribution, or near-duplicate detection across sections — and to enter it in the index alongside substance. BREADTH (§6) is a first step in that direction and is the reason we report it, but it is reported rather than incorporated. We implement no redundancy component and claim no result about one; this is recorded as the direction the objection actually points.

9.7 Specification adopted

Table 17 ranks $SUS^{\text{tfidf-}\ell}$ first, at 0.6817 against 0.6663. We adopt SUS^{dens} nonetheless, and the reason is not that the difference is negligible. The gap is 0.0154 between two measures that correlate 0.9926 with each other; whether it is statistically distinguishable is a question about dependent correlations that we do not test, so we neither claim the two specifications are equivalent on the criterion nor treat the ordering between them as informative. The criterion’s discriminating power in this section lies between the length-normalised pair and $SUS^{\text{tfidf-L2}}$ — a factor of two — and not within the pair.

The tie is therefore broken on a structural ground rather than an evidential one. SUS^{dens} estimates nothing on the sample: it requires no document frequencies, so a report’s score does not change when the corpus grows and is comparable across corpora, whereas both IDF-weighted specifications are defined relative to these 343 reports (§12). Between two specifications the convergent test does not separate, we take the one that does not carry the sample around with it.

Little rests on that choice and a great deal rests on the other one. The two length-normalised substance scores correlate 0.9926 and the labels they produce differ on 6 of 343 documents, figures §10 establishes; $SUS^{\text{tfidf-L2}}$ differs from the adopted specification on 96 of 343. §10 quantifies that consequence.

10 Robustness to the substance specification

§2 established that the substance component of Lagasio (2024) is underdetermined: the aggregation from per-term TF-IDF values to a document-level SUS_i is never stated, and the `norm` setting

is never mentioned. More than one implementation is therefore admissible on the published description. Of these two silences, this section takes up the second: the three specifications defined in §6 span the admissible choices of term weighting and vector normalisation. The aggregation silence is disposed of analytically rather than by comparison, in §7. This section measures how far the index and its labels move across the choices it does span.

Two things this section does not do. It does not choose a specification: the selection argument rests on agreement with an independently constructed yardstick and is made in §9. And it does not treat agreement between two specifications as evidence that either is correct — two estimators can agree closely and both be measuring the wrong quantity, which is precisely the diagnosis §7 reaches for one of the three. Throughout, the *score* normalisation is held fixed at *z*-scores; the sensitivity of the index to that separate choice is the object of §8.

10.1 Design

The three specifications cross the two open choices one factor at a time. SUS^{dens} and $SUS^{\text{tfidf-}\ell}$ share a denominator — document length — and differ only in whether terms are weighted uniformly or by IDF, so the pair isolates *term weighting*. $SUS^{\text{tfidf-}\ell}$ and $SUS^{\text{tfidf-}L2}$ share IDF weights and differ only in whether the term vector is divided by document length or scaled to unit L_2 norm, so that pair isolates *vector normalisation*. $SUS^{\text{tfidf-}L2}$ implements the inference — conceded as reasonable in §2 and §7 — that the four disclosed `TfidfVectorizer` settings were the only departures from library defaults, leaving the default L_2 setting in force; the original never states it in terms. We also report BREADTH, the effective number of distinct vocabulary terms (§6): a diagnostic rather than an index component, and a measure of topical spread that carries no information about how much ESG content a document holds.

10.2 Component-level sensitivity

Table 19 gives the correlation matrix over the three specifications and BREADTH.

Table 19: Pearson correlations among the three substance specifications and BREADTH ($n = 343$).

	SUS^{dens}	$SUS^{\text{tfidf-}\ell}$	$SUS^{\text{tfidf-}L2}$	BREADTH
SUS^{dens}	1.0000	0.9926	0.4177	0.4626
$SUS^{\text{tfidf-}\ell}$	0.9926	1.0000	0.4600	0.5118
$SUS^{\text{tfidf-}L2}$	0.4177	0.4600	1.0000	0.9733
BREADTH	0.4626	0.5118	0.9733	1.0000

The two contrasts can be read off directly. Across the weighting contrast, $r = 0.9926$: replacing uniform weights with IDF weights leaves the measurement essentially unchanged. Across the vector-normalisation contrast, $r = 0.4600$; between the two specifications that differ in both factors, $r = 0.4177$.

The BREADTH column identifies what the substitution does. $SUS^{\text{tfidf-}L2}$ correlates 0.9733 with a pure diversity statistic — higher than it correlates with either of the two alternative substance measures, and higher than either of those correlates with BREADTH. Under L_2 vector normalisation the score is closer to being a restatement of topical spread than it is to being a measure of substance. §7 supplies the mechanism: unit scaling discards vector magnitude and retains only direction, so what survives is the mix across the vocabulary, not the amount.

10.3 Index-level sensitivity and classification agreement

Table 20 carries the comparison through to the assembled index and its labels.

Table 20: Index correlation and classification agreement against ESGSI(SUS^{dens}) ($n = 343$). Labels follow the original’s rule, $\text{ESGSI} > 0 \Rightarrow$ “Potential ESG-washing”. Score normalisation is z -score in every row.

Substance specification	Factors varied	Pearson r	Flagged	Label flips
SUS^{dens} (reference)	—	1.0000	171/343	—
$\text{SUS}^{\text{tfidf-}\ell}$	weighting	0.9961	167/343	6 (1.7 %)
$\text{SUS}^{\text{tfidf-L2}}$	weighting, vector norm.	0.6964	159/343	96 (28.0 %)

$\text{SUS}^{\text{tfidf-L2}}$ differs from the reference in both factors, so its 96 flips are not, by themselves, evidence about which factor drives them; the attribution needs an argument, not a reading of the table. The argument is a bound, and it uses only the two counts already reported. At most 6 documents anywhere in the corpus disagree under weighting alone (the $\text{SUS}^{\text{dens}}\text{--}\text{SUS}^{\text{tfidf-}\ell}$ row). Of the 96 documents on which SUS^{dens} and $\text{SUS}^{\text{tfidf-L2}}$ disagree, therefore, at most 6 can be documents that already disagreed under weighting before vector normalisation was even applied. At least $96 - 6 = 90$ of the 96 are documents on which weighting left the label unchanged and vector normalisation alone flipped it — for those documents, vector normalisation is the entire explanation, with weighting contributing nothing. Vector normalisation accounts for at least 90 of the 96 combined label changes; weighting cannot be credited with more than the 6 flips it produces on its own, anywhere in the corpus. The ranking in Table 20 is not an artefact of comparing every specification to SUS^{dens} rather than to one another: vector normalisation dominates under any attribution consistent with the two counts already in the table.

Separately, every row of Table 20 shares the same $Z(\text{SEN})$ term, and SEN is only weakly related to any of the three substance measures. Subtracting a common component that carries little of its own relationship to the compared variables tends to inflate the correlation between two differences relative to the correlation between the terms being subtracted — not as a matter of mathematical necessity, but empirically, and it does so here: every index-level correlation in Table 20 runs above its component-level counterpart in Table 19 (0.9961 against 0.9926; 0.6964 against 0.4177). Read the index-level number accordingly: it is not a tighter estimate of how closely the underlying substance measurements agree than the component-level number is. If anything it is the looser of the two, inflated by a term the compared specifications do not even differ on.

10.4 Interpretation

The result is one sentence: *term weighting is second-order and vector normalisation is first-order*. Substituting uniform weights for IDF moves six documents out of 343; changing the vector normalisation moves ninety-six. Roughly a quarter of all classifications depend on a parameter the original never states in terms. What that establishes is a matter of stakes rather than of bookkeeping, and it should be read alongside the concession §7 makes: the **norm** setting is recoverable from the convention that only non-default arguments are reported, and we rely on that convention ourselves, so we do not present the silence as a reproducibility failure. We present it as a first-order choice left to an inference. The two implementations the published text does not distinguish *explicitly* disagree about the label of more than one report in four, which is why we report both rather than settle the question by convention.

The aggregation rule is the opposite case in both respects, and §7 disposes of it by argument rather than by comparison: no convention supplies a default for it, yet because the index is z -scored a positive affine rescaling such as switching from a sum to a mean over the vocabulary cancels out, so that silence does not move a single label even though nothing in the original resolves it.

It follows that the near-identity of SUS^{dens} and $\text{SUS}^{\text{tfidf-}\ell}$ is not a validation of either. It

establishes insensitivity to one parameter, which is a property of the parameter, not a warrant for the measure. Both could be poor measures of substance and would still agree at 0.9926. What separates the specifications is external evidence, and that is §9’s subject; the present section only quantifies the cost of getting the choice wrong.

10.5 Robustness to the vocabulary

The section so far varies how the vocabulary is weighted and how the resulting vector is normalised, never which terms it contains. That leaves untouched the standing objection to any dictionary-based measure: the result is an artefact of the word list, and a different list would have produced a different index. §5 puts that objection in measured form — an external audit of 880 zones from four FY2023 reports identified 147 ESG terms present in the text and absent from our 154 entries — and those terms are the material for a robustness check on the vocabulary itself.

Each of the 143 distinct candidate strings behind that list was measured across all 343 reports for frequency and dispersion and classified on those statistics (§5); admitting the dispersed entries and the polysemous ones handled by collocation, and holding back those flagged sectoral, takes the vocabulary from 154 to 289 lemmatised TF-IDF entries. Table 21 varies it in two steps: the expansion, then the 22 terms held back as sectoral.

Both experiments run on the same 343 documents and score them with different vocabularies; the extracted text is held fixed at the expansion’s own extraction. That isolation is deliberate and it is not free: our vocabulary file drives the paragraph-selection regex of §4 as well as the TF-IDF vocabulary, so an end-to-end comparison of the two vocabularies would confound what is scored with what is extracted. Holding the corpus fixed removes the confound at the cost that the first column of Table 21 is not the index reported elsewhere in this paper: the adopted specification of §5 scores its own extraction, where mean SUS^{dens} is 5.897 rather than 5.168. The difference between those two figures is an extraction effect, and §12 reports it.

Table 21: Sensitivity of the index to the scoring vocabulary, corpus held fixed ($n = 343$). Each column after the first is compared with the column immediately to its left. All three columns score the same extracted text, so the first is the adopted 154-term vocabulary on the expansion’s extraction and not the run reported elsewhere in this paper (see text). Labels follow the original’s rule, $ESGSI > 0 \Rightarrow$ “Potential ESG-washing”; score normalisation is z -score in every column. The trend is negative and significant beyond any conventional level in all three ($p < 10^{-7}$ throughout), with the independence caveat of §11 attaching to each.

	154 terms	289 terms	289 + 22 sectoral
Mean SUS^{dens}	5.168	6.516	6.541
Change vs preceding column	—	+26.1 %	+0.4 %
Flagged	172/343	164/343	167/343
Label flips vs preceding column	—	18 (5.2 %)	3 (0.9 %)
Pearson r vs preceding column	—	0.9882	0.9998
$ESGSI(SUS^{\text{dens}})$ slope	−0.219/yr	−0.194/yr	−0.193/yr

Near-doubling the vocabulary raises mean substance density by 26.1 %, so the added terms match text rather than nothing, and it moves the index very little: $r = 0.9882$, 18 of 343 labels change (5.2 %), and the flagged count falls from 172/343 to 164/343. The temporal slope of §11 goes from −0.219 to −0.194 per year on this corpus (§11.6). The 22 terms held back as sectoral move 3 labels of 343 when added on top of the expansion; §12 reports what they do at firm level and why they are excluded.

Two limits bound the check narrowly. The 289-term list is not independent of the corpus it is tested on: its candidates came from auditing four of these reports and were classified on statistics computed from all 343. And the check is one-sided, because terms were admitted for being broadly dispersed, which is precisely the property that makes a term least able to move a density ranking; a vocabulary built on other principles is untested, and nothing here bounds what one would do. What the check bounds is the index’s sensitivity to a defensible extension of this vocabulary on this corpus. It does not establish that the vocabulary is right.

The check is not specific to the terms we chose. Deleting 30 of the 154 entries at random and rescoreing — 5,000 draws — moves a median of 17 labels at a mean index correlation of 0.988; our 18 flips and 0.9882 sit at the 56th and 58th percentiles of that distribution, and among the random draws that displace the index as far as ours does the median is 19 labels against our 18. What the experiment measures is therefore the index’s insensitivity to a vocabulary perturbation of this magnitude, not the adequacy of the particular terms admitted. Matched on occurrence mass rather than on term count the curated expansion is modestly the more stable of the two, at the eighteenth percentile of the random distribution; we report that without leaning on it.

Part of the agreement is structural rather than empirical. The 154 entries are retained whole inside the 289, so the increment is additive, and its standard deviation is 0.236 of the base measure’s. That ratio alone bounds the correlation between the two substance scores below at $\sqrt{1 - 0.236^2} = 0.972$ and the correlation between the two indices at 0.984. The observed 0.9882 lies about a quarter of the way up a window 0.016 wide. The exercise is not vacuous — a fifth of the expanded numerator is new material — but the range in which the answer could have landed was fixed by the acceptance rule before any document was scored, because terms admitted on dispersion contribute a near-constant per-document increment.

Placed beside Table 20, the two experiments complete the set of choices this paper varies (Table 22).

Table 22: Labels moved by each of the three choices this paper varies ($n = 343$; z -score normalisation and the ESGSI > 0 threshold throughout). Each row names its own baseline: the third is an increment on the robustness vocabulary, not on the adopted one. The rows are not commensurable perturbations and the column does not rank the choices by importance; see the text.

Choice varied	Measured against	Labels moved
Vector normalisation (Table 20)	ESGSI(SUS ^{dens}), 154 terms	96/343 (28.0 %)
Scoring vocabulary, 154 $\rightarrow$ 289 terms	ESGSI(SUS ^{dens}), 154 terms	18/343 (5.2 %)
Sectoral terms, +22	the 289-term vocabulary	3/343 (0.9 %)

The three rows are not perturbations of comparable size, and the column should not be read as ranking the choices by how much each matters. Across 2,354 random perturbations of this vocabulary, label flips scale as $(1 - r)^{0.57}$, close to the square root of the index displacement. On that scaling the three rows differ by a factor of 1.3 although the displacements behind them differ by a factor of 1,500: each moves about as many labels as any perturbation of its size would move. What separates the rows is how far each moved the index, not what kind of choice each was. Nor is the third row’s margin over the second what it appears: measured against the adopted specification rather than incrementally, the full 289 + 22 vocabulary moves 17 labels, not 3. What the column licenses is narrower than a hierarchy and is still this section’s point: of the three choices we varied, the one Lagasio (2024) never states in terms is the one that changes the most classifications.

The specification adopted for the paper is unchanged — the 154 entries of §5, on which every figure reported elsewhere is computed. Both vocabularies here are robustness instruments, not candidates for re-selection: the selection argument is §9’s, and it is not re-run on the expanded list.

10.6 The threshold qualifies the flip counts

Label flips are counts of documents crossing $\text{ESGSI} = 0$, and that boundary is definitional. The original states the rule and reports no calibration of it against any external benchmark (§6). With score normalisation fixed at z -score throughout this section, the consequence is structural: ESGSI is a difference of two mean-zero variables, so it has mean zero by construction and the > 0 rule flags close to half the corpus under every specification — 171, 167 and 159 of 343 reports respectively. The cut-off is a property of the sample, not of any firm in it.

Two implications. First, the flip counts are boundary-dependent: imperfectly correlated indices disagree first about documents lying near the cut-off, so a threshold placed elsewhere would return different counts. The absolute flip rate is not an error rate and should not be read as one. Second, the comparison between the two rows is unaffected, because both are measured against the same boundary on the same sample: whatever the threshold does to 6, it does to 96. The contrast between the two factors is what this section claims; the level is conditional on a convention. §12 takes up the arbitrariness of that convention, and of the relative ranking it rests on, as a limitation of our own index and not only of the original’s.

Finally, the same underdetermination applies to any result computed from this index. The temporal series in §11 is therefore reported under all three specifications rather than under the adopted one alone.

11 Temporal results

The original is cross-sectional by construction, and its conclusion names the extension it does not attempt: “The cross-sectional nature of our data limits causal inferences, highlighting the need for longitudinal studies to examine how changes in corporate characteristics and regulatory environments affect ESGSI over time” (Lagasio, 2024, p. 10). This section supplies that series on our panel.

It is reported last among the empirical sections, and it is subordinate to what precedes it. §7 establishes that a TF-IDF substance score measures the *mix* of ESG terms rather than their amount, and §8 that the original’s headline conclusion follows from an undisclosed score-normalisation choice. Having argued that the instrument is unsound under one admissible specification, we cannot then present a time series of it as the paper’s contribution. What follows is the payload the corrected instrument delivers, conditional on the choices justified in §9 and bounded by the sensitivity documented in §10. The central claim remains §8’s. By the end of the section the reason for the subordination will be sharper than a matter of ordering: §11.4 identifies a mechanism under which the series would move exactly as it does with no change in firm conduct whatever, and that mechanism is a property of the index architecture rather than of our panel.

The series is computed under $\text{ESGSI}(\text{SUS}^{\text{dens}})$, the specification §9 selects on convergent validity against QUANT. We report the base index only: the extended index carries two arbitrary weights, and §12 sets out why its temporal slope is contaminated rather than cleaner.

11.1 The series

Table 23 gives the yearly mean. The panel is complete: 49 firms, seven years, 343 documents, exactly 49 in every year, with no entry, exit or backfill. The mean falls from 0.563 in 2018 to -0.674 in 2024 and is strictly decreasing in every year of the window. An OLS regression of the document-level index on calendar year ($n = 343$) gives a slope of -0.205 per year with $p = 1.7 \times 10^{-8}$.

Three qualifications attach to that regression before anything is read off it.

The slope is a summary, not a model of the path: the year-on-year decrements are uneven, and nothing here identifies a constant annual rate. The crossing from positive to negative between

Table 23: Mean ESGSI by year, under the adopted substance specification and under the L_2 -normalised alternative. Balanced panel, 49 documents per year. Values are read from the pipeline output, which standardises with the population standard deviation.

	2018	2019	2020	2021	2022	2023	2024
ESGSI(SUS ^{dens}) (adopted)	0.563	0.471	0.161	0.007	-0.121	-0.407	-0.674
ESGSI(SUS ^{tfidf-L2})	0.561	0.453	0.106	-0.168	-0.274	-0.371	-0.307
Documents	49	49	49	49	49	49	49

2021 and 2022 is likewise not a calibrated event. Both components are score-normalised across the pooled sample, so the index has mean zero over the seven years by construction; the location of the zero crossing is a property of the estimation window, not of the disclosure. The slope is invariant to that centring; the levels are not.

The p -values reported throughout this section, here and in Table 26, come from OLS on document-level observations and treat the 343 documents as independent. They are not. The panel is 49 firms observed seven times each, and both firm-level heterogeneity and within-firm serial correlation are present by construction, so the effective number of independent units is nearer 49 than 343 and the conventional standard errors are anti-conservative. We do not report clustered or panel-corrected inference. The magnitudes that survive that objection are not the p -values but the facts they summarise: seven consecutive yearly means falling in sequence, a decline of comparable sign and larger magnitude reappearing in a disjoint single-country subsample, and driver quantities that are directly counted rather than estimated (Table 24). Readers who discount the exponents should discount them; the section’s argument does not rest on any of them being exact.

11.2 What moves, and what does not

ESGSI = $Z(\text{SEN}) - Z(\text{SUS})$, so a decline is either tone falling or substance rising. The displacement sits on the substance side. Table 24 decomposes the endpoints.

Table 24: Driver decomposition, 2018 against 2024. Per-report means.

Quantity	2018	2024	Change
Extracted ESG text per report (lemmatised tokens)	27,906	63,654	+128.1%
ESG mentions per report	1,382.7	4,832.0	+249.5%
SUS ^{dens} (mentions per 100 words)	4.87	7.45	+52.8%
BREADTH (effective distinct terms)	28.99	41.73	+43.9%
Extracted zones per report	178.5	275.2	+54.2%

Three quantities grow together. The volume of text the extraction stage identifies as ESG-relevant more than doubles; within that text, ESG mentions grow faster still, so density rises by half again against the larger denominator; and the mentions spread over more of the vocabulary, raising the effective number of distinct terms by 44%. Reports are not merely longer. They are lexically denser in ESG terms and broader in coverage.

SEN shows no detectable trend. Regressed on calendar year over the same panel, its slope is +0.0016 per year with $p = 0.666$: a point estimate two orders of magnitude below the index slope, and of the wrong sign to contribute to the decline, since rising tone raises ESGSI. Two things follow and a third does not. It follows that the movement in the index is located on the substance side, and that the decline is not produced by firms writing in a less positive register. It does not follow that tone is uninformative, or that SEN measures nothing: a failure to reject at $p = 0.666$ is an absence of detectable movement, not a demonstrated null, and SEN retains

cross-sectional variance that enters every index value in the panel. We claim the location of the movement, not the irrelevance of the component.

The null does dispose of one specific objection, and only that one. A dictionary-based index applied longitudinally invites the charge that a fixed word list scores 2024 language differently from 2018 language, so that an apparent trend is dictionary drift rather than disclosure change. For the *sentiment* dictionary that charge would bind only if the movement were in SEN, and it is not. It does not extend to the ESG vocabulary, which is where the load is carried and where the equivalent objection is live rather than closed: §11.4 takes it up, and does not resolve it.

What the series does not establish is that the added substance is real activity. The index registers that firms wrote more, and more diversely, about ESG topics per unit of text. Whether that text corresponds to conduct is outside what any text-derived index can determine.

11.3 Alternative explanations

Table 25 lists the explanations a downward trend in this index admits besides a change in disclosure, the check available against each, and what that check settles. Two rows are not settled: the first of the two is bounded but not removed, the second is neither, and both are the subject of §11.4.

Table 25: Alternative explanations for the decline, and the status of each after the checks this design supports.

Explanation	Check	Status
Country and document-type mix drift across years	Balanced panel: 49 documents and a fixed per-country count every year; 33–36 annual reports, 12–15 URDs, 1 sustainability report per year (§3)	Excluded by construction
The country/type confound generates the slope	Germany: 98 documents, one country, one document type, $-0.301/\text{yr}$	Excluded within that subsample
Reports simply got longer	SUS^{dens} is a ratio; dilution lowers it (§7)	Excluded; biases against the finding
Extraction admits more text over time	Marginal zones enter at the density threshold, below the mean; extraction-stage density rises across the window	Excluded on the extensive margin
ESG language concentrated rather than increased	None: no whole-document density is computed by year	Not testable in this design
Sentiment dictionary scores 2024 language differently	SEN slope $+0.0016/\text{yr}$, $p = 0.666$	No movement to explain
ESG vocabulary contains entries whose referents postdate 2018	Recomputed with the 34-term reporting-frameworks block deleted: slope $-0.155/\text{yr}$, $p = 2.5 \times 10^{-5}$ (§11.4)	Bounded, not excluded
Mandated disclosure raises term counts without a change in conduct	None: exposure is common to all firms and collinear with calendar time	Not excluded (§11.4)

Composition. The panel is exactly balanced. The same 49 firms appear in all seven years, every year contributes exactly 49 documents, and every country’s own count is identical in every year (§3). The country mix cannot drift because there is nothing for it to drift into. Document-type composition is likewise stable: every year contains between 33 and 36 annual reports, between 12 and 15 URDs and exactly one standalone sustainability report, a year-to-year variation of at most three documents in 49 and not directional.

Balance is bought at a price that belongs here rather than in a later section. Requiring one report per firm in each of seven consecutive years selects firms that reported continuously across the window and survived it. The trend is therefore a within-firm trend among continuous reporters, and the sample it generalises to is that population, not European issuers at large. This restricts external validity; it does not generate an internal trend, since the selection is applied identically to every year.

Length. A report that grows without adding ESG content does not raise SUS^{dens} ; it lowers it. §7 demonstrates the point directly: appending 56,000 non-ESG filler words to a 22,576-word document, with ESG mentions held at 695, drops density from 3.0785 to 0.8845 per 100 words, while the L_2 -normalised score does not move at all. The denominator is doing exactly what a density denominator should. Length growth on its own is therefore not a rival explanation for the decline in ESGSI but a force working against it, and the observed rise in density is net of whatever dilution the growth in report length produced.

Extraction yield. SUS^{dens} counts vocabulary hits over the lemmatised tokens of the *extracted* zones, not of the whole document, and the extraction stage admits more text in later years — zones per report rise from 178.5 to 275.2. If the measure were mechanically tracking that yield, the growth would be an artefact of the pipeline. It is not, and the direction of the selection is why. Zones are built around paragraphs that clear a keyword-density threshold of 1.0 per 100 words, with one paragraph of context admitted on either side (§4). Text admitted at the extensive margin therefore enters at or near that threshold, well below the corpus mean, and the context windows import adjacent text with no density requirement at all: adding zones dilutes. The extraction-stage keyword density — the §4 quantity, computed with the full regex over raw text, and not to be read as SUS^{dens} — nevertheless rises monotonically across the window rather than falling, and it does so under both readings of that quantity, the pooled ratio and the mean of per-report values. The intensive margin dominates the extensive one, which is the opposite of what a yield artefact would produce.

One residual is not disposed of. Extraction and measurement share a vocabulary (§4), so density is measured on text selected for being ESG-dense. A firm that held its total ESG content fixed but consolidated it into denser, more clearly delimited passages — a dedicated non-financial statement in place of remarks scattered through the report — would raise the density of the extracted zones without raising the density of the document. No whole-document density is computed by year, so we cannot separate that from an increase in content, and the yield of the extraction stage relative to the full PDF was measured on a 20-document sample rather than year by year. We flag this rather than dismiss it.

The single-country subsample. The strongest of the composition checks is that the trend holds inside one country with one document type. Germany contributes 98 documents — 14 firms over seven years — and all 98 are annual reports. Within that subsample the slope is -0.301 per year with $p = 1.2 \times 10^{-6}$, steeper than the pooled estimate rather than attenuated (Table 26).

Table 26: Trend in ESGSI(SUS^{dens}), pooled panel and single-country subsample. OLS of the document-level index on calendar year. Both p -values assume independent observations, which 49 firms observed seven times each are not.

Sample	n	Slope/yr	p	Composition
Full panel	343	-0.205	1.7×10^{-8}	7 countries, 3 document types
Germany only	98	-0.301	1.2×10^{-6}	1 country, 1 document type

What this check removes, and what it does not, must be stated exactly. §3 identifies a near-total confound between country and document type and treats any cross-sectional comparison along either dimension as uninterpretable. Inside the German subsample neither factor varies, so neither can be generating the slope *there*. That is the whole of the inference. It does not make the German subsample representative, and it does not transfer to the pooled estimate by extension: it shows that a trend of this sign and magnitude exists in a slice where the confound is absent, not that the confound is absent from the pooled slope. The German slope is in fact steeper than the pooled one, which is itself evidence that the rate is heterogeneous across the panel, so -0.301 is not to be read as a European rate. The subsample is 98 documents from 14 firms, so the dependence between observations noted for the pooled regression binds harder here, not less. And it is silent by construction about anything acting on all seven countries at once — which is the next subsection’s problem, and the one this check is least able to address.

11.4 The objection this design cannot answer

The window 2018–2024 is the interval over which European non-financial reporting moved from largely voluntary to substantially prescribed. A firm that adds a taxonomy-alignment table, a framework index and a standards-referenced non-financial statement adds ESG vocabulary to its report in quantity, without necessarily changing anything it does. Our numerator counts words. It cannot distinguish the two, and the objection is the strongest available against reading this series as evidence about firms.

The mechanism is not hypothetical in our instrument; it is visible in the vocabulary. The 154-term list contains entries — *csrd*, *esrs*, *issb*, *sfdr*, *tcfd*, and four members of the *taxonomy* family — that name regimes, standards and disclosure categories which either did not exist or were not in force at the start of the window. Their near-absence from 2018 reports is a matter of chronology, not of disclosure behaviour. A fixed word list applied across a period in which the referents of some of its entries came into being will register an increase that is to that extent definitional. This is the ESG-vocabulary counterpart of the dictionary-drift objection that the SEN null disposes of on the sentiment side. Here it is not disposed of. The definitional part of it can be bounded, and below we bound it; the objection as a whole survives that bound.

§9 supplies one measurement of the drift, and it points the wrong way for us. Fourteen vocabulary entries overlap the QUANT regex — eleven in the **frameworks** group, three in **units** — and carry 7.50% of all SUS occurrences pooled. That share is not stationary: the **frameworks** group rises from 10.15% of all QUANT hits in 2018 to 21.87% in 2024. Framework and standard references roughly double as a share of quantified disclosure across exactly the window in which the series declines. The drift is real, it is measured, and it is in the direction the objection predicts.

Two counterweights are available, and neither disposes of the confound; the first bounds the part of it that is purely definitional, the second is an argument only.

The first is to delete the offending vocabulary and recompute. The *reporting frameworks and ratings* block of §5 is the block in which regime names live, and it is where every entry above sits. Dropping it in its entirety — all 34 of its terms, taking the vocabulary from 154 to 120 — removes far more than the chronologically suspect entries, since the block also contains framework vocabulary that predates the window. It is deliberately over-inclusive: we want an upper bound on what the definitional component can be worth, so we discard the whole block rather than adjudicate entry by entry. Table 27 reports the result. The rise in density falls from +52.8% to roughly +43%, and the index slope from -0.205 to -0.155 per year — about three quarters of its magnitude — remaining significant at any conventional level.

What that check bounds and what it leaves untouched must be stated exactly, because it is easy to overread. It shows that the trend is not an artefact of counting regime names: strip every one of them, and three quarters of the slope is still there. It does *not* dispose of the mandate confound, and we do not offer it as doing so. A reporting regime elicits substantive disclosure

Table 27: Temporal trend with the *reporting frameworks and ratings* block removed from the ESG vocabulary (154 $\rightarrow$ 120 terms). Density is the mean of per-report SUS^{dens} values. p -values carry the same independence caveat as everywhere else in this section.

	Full vocabulary (154)	Frameworks excluded (120)
Mean density, 2018 $\rightarrow$ 2024	4.87 $\rightarrow$ 7.45	4.43 $\rightarrow$ 6.35
ESGSI(SUS^{dens}) slope	-0.205/yr	-0.155/yr
p	1.7×10^{-8}	2.5×10^{-5}

as well as its own acronyms — emissions-scope terms, net-zero language and the greater part of the environmental vocabulary are prescribed by the same regimes without naming them — and that portion of the effect is not separable from conduct by any vocabulary edit, because the words involved are exactly the words a firm that had genuinely changed its conduct would also use. The 34-term deletion bounds the purely definitional component of the confound. The rest of the confound is untouched, and it is the larger part.

The second counterweight is breadth. A compliance template inserts a bounded set of named items, which predicts more mentions concentrated on few terms. What we observe is broadening: the distinct vocabulary entries present per report rise from 68.65 to 105.45 of 154, and BREADTH from 28.99 to 41.73. That is more than a handful of new acronyms can produce. It is not decisive either, because the mandates in question prescribe reporting across environmental, social and governance topics simultaneously, so a mandate effect can broaden coverage as well as deepen it. The counterweight restates the confound at a higher level rather than resolving it.

No design in this paper can do better than that bound, and it is worth being precise about why the residue is not reachable. A reporting mandate is a common time shock. It applies to all 49 firms in the same years, so its exposure is collinear with calendar year, which is the regressor. The balanced panel, the document-type check and the German subsample all remove *cross-sectional* composition; none of them removes anything that moves with time. The German result is the weakest defence available here rather than the strongest, since German issuers face the same European regimes as the rest of the panel: holding country and document type fixed does nothing to hold the regulatory calendar fixed. Identification would require variation in exposure — firms inside and outside scope, or staggered application dates — which a panel of 49 large European issuers, all in scope, does not contain.

The consequence is a restriction on the claim rather than a hedge around it. What the series identifies is that ESG-lexical content per unit of extracted text rose across 2018–2024 while tone did not. Whether the added text reflects changed conduct, compliance with a new reporting regime, or unchanged conduct newly described in a prescribed vocabulary is not identified here, and the second and third of those are the ones an informed reader should regard as most likely to be doing work.

There is a sharper consequence, and it belongs to §8’s argument rather than to this section’s. Under the mandate reading the index does not merely fail to observe conduct. $ESGSI = Z(\text{SEN}) - Z(\text{SUS})$ falls when substance rises, so a firm that adds a mandated taxonomy table is scored as less likely to be greenwashing — by construction, for having complied. An instrument whose washing score is mechanically reduced by regulatory compliance is not measuring the discrepancy between talk and action; over a window of tightening mandates it is measuring the volume of prescribed talk. That is a property of the architecture we inherit, not of our corpus, and it applies to any application of this index family across a period of regulatory change, the original’s included. The series is accordingly better read as a further demonstration of §8’s point than as a finding about European firms.

11.5 Sensitivity of the series to the substance specification

The direction and the significance of the trend hold under all three substance specifications of §6: the yearly means decline and the slope is negative at conventional levels under SUS^{dens} , $SUS^{\text{tfidf-}\ell}$ and $SUS^{\text{tfidf-L2}}$ alike.

The shape does not hold uniformly, and we state the difference rather than smooth it. The yearly series is strictly monotone under SUS^{dens} and under $SUS^{\text{tfidf-}\ell}$. It is *not* monotone under $SUS^{\text{tfidf-L2}}$: 2024 rebounds to -0.307 , above the 2023 value of -0.371 (Table 23), so that specification yields a series that is monotone over 2018–2023 and reverses in the final year. Monotonicity is therefore a claim about our specification and about the length-normalised TF-IDF variant, and we make it only for those two; what generalises across all three is the sign and the significance of the slope.

This precision is required rather than fastidious. §10 has just established that the three specifications disagree on 28.0% of the labels and that the choice between them is not settled by anything the original discloses. A temporal section that then asserted uniform behaviour across those same specifications would be claiming a stability that the preceding section demonstrably refutes. The rebound is also the one place in the window where the two series separate visibly, and it is consistent with §10’s diagnosis: $SUS^{\text{tfidf-L2}}$ correlates 0.9733 with BREADTH, so a year in which coverage broadens without a commensurate rise in amount moves it differently from a density measure.

11.6 Sensitivity of the series to the vocabulary

A dictionary index invites one further objection of the same family as the composition ones: that the decline is a property of the word list rather than of the reports. The check is to move the boundary of the list and recompute. The audit of §5 supplies the material — 147 ESG terms present in the extracted text and absent from our vocabulary — and each candidate was measured across all 343 documents for frequency and dispersion before admission, taking the vocabulary from 154 entries to 289.

Under the 289-entry vocabulary the yearly series is essentially unchanged. The slope moves from -0.219 to -0.194 per year, each significant at $p < 10^{-7}$, and restoring the 22 entries the same exercise had set aside as sector-specific leaves it at -0.193 . Both figures are computed on the expansion’s extraction with the corpus held fixed (§10.5), which is why the base slope here is -0.219 rather than the -0.205 of Table 23: the two are not differenced, and only the proportional movement between them is read. The series does turn on where the boundary is drawn, and this section contains both directions. Moving it outward, by the 135 audited entries, changes the slope by about a ninth; moving it inward, by striking the 34-term reporting-frameworks block (Table 27), changes it by about a fifth, and the two are of the same order rather than of different ones. What the outward check bounds is the sensitivity of the series to a defensible extension of this vocabulary, built from these documents and admitted on dispersion measured over them. It bounds nothing about a vocabulary built on other principles.

This is not a rebuttal of §11.4, and the two must not be run together. Vocabulary insensitivity is a statement about our discretion: had we drawn the boundary of this word list where the audit supports drawing it instead, the trend would be what it is, so the result is not an artefact of that choice. The mandate confound is a statement about chronology: some entries name regimes that did not exist in 2018, and could not have appeared in an early report whatever the firm was doing. Enlarging the vocabulary with further contemporary ESG terms does nothing about the second, and there is reason to think it aggravates it. The candidates were harvested from FY2023 reports (§5), so the added entries are drawn from language at the end of the window; every framework name in the 154-term list is carried forward untouched; and the enlarged list is not independent of the corpus it is tested on. The bound on the definitional component is still the one in Table 27 — -0.155 per year with the reporting-frameworks block struck out — and

the vocabulary check neither tightens nor loosens it. Robustness to the lexicon and identification of what the lexicon counts are different questions. The first is bounded here, in one direction, for a class of terms selected on this corpus for the property that makes them least able to move a density ranking; the second is not settled anywhere in this paper.

11.7 What the series licenses

Both components enter the index after score normalisation across the sample (§8’s object, not §7’s), and one consequence must be explicit before any of these numbers is quoted elsewhere. The index has no cardinal meaning. A mean of -0.674 is not a quantity of ESG-washing but a position within the distribution of this corpus. Levels are not comparable between corpora: no level reported here is comparable with any figure in Lagasio (2024), or in any other application of the same architecture, whatever the score-normalisation rule. This holds even though SUS^{dens} is the one substance specification whose raw value is corpus-independent, because the score normalisation reintroduces sample dependence downstream, at index assembly. The slope inherits the same property: -0.205 per year is in pooled standard deviations of this sample per year, so it is no more transferable than the levels are, and a slope estimated on another corpus would not be commensurable with it. Only the within-sample ordering and the direction of change over time are interpretable.

Within those bounds, the result is this. On a balanced panel of 49 continuously reporting European firms over 2018–2024, mean tone-minus-substance declines strictly, and the decline is carried by the substance component, with no detectable movement in tone. It is not a composition artefact: the panel is balanced by construction, and the decline survives in a subsample where the corpus’s country and document-type confound does not exist. It is not a length or an extraction-yield artefact, both of which would bias the measure the other way. Its sign and significance are invariant to the substance specification, though its monotonicity is not, and they survive at roughly three quarters of magnitude when every reporting-framework term is struck from the vocabulary. What it is not shown to be is evidence about conduct, and the reporting mandates that arrived over the same window supply a specific mechanism by which the index would move as it does with conduct held fixed (§11.4) — a mechanism whose purely definitional part we can bound but whose remainder no design available to us can separate from calendar time. That is what the corrected instrument yields over time: a demonstration of consequences, not the paper’s claim.

12 Limitations

A paper arguing that a published conclusion is an artefact of undisclosed choices is held to its own standard, so the accounting below is complete rather than selective. (The original’s runs to two paragraphs in its conclusion and acknowledges two items: the cross-sectional design, and that ESGSI measures could be more sophisticated, p. 10.) Table 28 attributes each limitation to our design or to the index we re-implement, descriptively only. Inheritance is not a defence: an inherited defect constrains our results exactly as much as one of our own, and the first two items apply to Lagasio (2024), and to any future user of this index family, with equal force.

12.1 The threshold is definitional, and the index has no cardinal scale

Under score normalisation by z -score, $ESGSI = Z(\text{SEN}) - Z(\text{SUS})$ is a difference of two mean-zero variables, so the rule $ESGSI > 0 \Rightarrow$ “Potential ESG-washing” partitions any sample at approximately its median by construction. Our 171 flagged documents of 343 are arithmetic, not an estimate: no figure in this paper estimates how much ESG-washing this corpus contains, and no reader should take one as such. The threshold is inherited. The original adopts the same zero

cut with no calibration against any external case and never reports the share it flags, although its own published quartiles bound that share at no more than 25 % of 749 (§2).

Both components are rescaled across the sample, so neither they nor the index carry cardinal meaning. An ESGSI of +1.5 is a position in this corpus’s distribution, not a quantity of anything; values are not comparable across corpora, across vocabularies, or between this study and the original. Only within-sample ordering, and change over time with composition held fixed, are interpretable.

12.2 Corpus dependence

$SUS^{\text{tfidf-}\ell}$ and $SUS^{\text{tfidf-L2}}$ estimate document frequencies on the analysed sample. Adding documents changes every idf and hence every historical score, so a report’s 2018 value is not stable under corpus extension — a live defect for a longitudinal design, which must re-score the whole panel whenever it grows. SUS^{dens} estimates no document frequencies, so its substance component is stable under extension; that does not lift the limitation from the index. Score normalisation is cross-sectional under every specification, so every ESGSI reported here, including the whole yearly series of §11, would move if the panel were extended. The series is one of positions within this corpus, not of levels.

12.3 Confounds and unjustified constants

Country and document type are near-perfectly confounded (§3): 99 of 101 URDs are French, all 98 German documents are annual reports, and all seven standalone sustainability reports are French. No cross-country or cross-type comparison in this corpus is interpretable and we make none; the country and type means in §3 identify neither effect. The within-Germany analysis (§11) holds both fixed — one country, one document type, 98 documents, slope $-0.301/\text{yr}$, $p = 1.2 \times 10^{-6}$ — but it does so on 14 firms, and it establishes that the decline is not produced by composition, not that the pooled magnitude is right. The cross-section is 49 firms throughout, observed seven times each, so the p -values in §11 are anti-conservative for the reasons given there; we report no clustered or panel-corrected inference.

The extended-index weights $w_q = w_h = 0.5$ were chosen without justification and never varied. We report no result that depends on $ESGSI^{\text{ext}}$, which is why no sensitivity analysis was run — not because the choice is defensible, and any use of $ESGSI^{\text{ext}}$ beyond this paper would need one.

12.4 QUANT is not independent of what it adjudicates

QUANT is the criterion on which the specification choice rests. The selection of SUS^{dens} over $SUS^{\text{tfidf-L2}}$ in §9 turns on their agreement with QUANT, and the argument is a convergent-validity argument: it is worth what the independence of the two instruments is worth. That independence fails in two ways, one of which we can remove by recomputation and one of which we cannot. Fourteen entries of the 154-term ESG vocabulary are also counted by QUANT — eleven by its **frameworks** group and three (*co2*, *co2e*, *ghg*) by its **units** group — and they carry 7.50 % of all SUS occurrences. The two groups that contain the shared terms are together 36.77 % of all QUANT hits (**frameworks** 17.22 %, **units** 19.55 %), so the contamination is not confined to a corner of the criterion. Nor is it stationary: the **frameworks** share rises from 10.15 % of QUANT hits in 2018 to 21.87 % in 2024, so the shared component grows across exactly the window over which §11 reports a trend.

Two consequences follow, and they are different in kind. The first is settled by recomputation rather than by argument, and §9.3 reports it: stripping the 14 shared entries from the vocabulary and QUANT of the **frameworks** and **units** groups drops every correlation by roughly two fifths — SUS^{dens} from 0.6663 to 0.4112, $SUS^{\text{tfidf-}\ell}$ from 0.6817 to 0.4125, $SUS^{\text{tfidf-L2}}$ from 0.3217 to

0.1799. So the levels reported there are upper bounds and the true level of convergent validity is moderate, not strong; that limitation stands and we state it in those terms. The ranking is not similarly affected — the length-normalised specifications stay roughly twice as close to QUANT as $SUS^{\text{tfidf-}L^2}$, and the ratio widens slightly, $2.29\times$ against $2.07\times$. We had previously nothing better than an argument for why the ranking should survive, and that argument does not hold (§9); what supports it is the recomputation and nothing else. The second consequence touches the extended index: ESGSI itself does not contain QUANT and is unaffected, but $ESGSI^{\text{ext}}$ subtracts both $Z(\text{QUANT})$ and $Z(\text{SUS})$, so there the shared mass reinforces rather than cancels and the growing **frameworks** share inflates the extended index’s temporal slope.

What recomputation does not touch is the more serious of the two failures, and it is the one that remains the most serious limitation in this paper. QUANT and SUS are computed over the same extracted zones, with denominators that are two measures of the same document’s length (§9). Deleting shared strings does not make the two instruments independent samples of anything; both still see only the text §4 admitted, and no vocabulary edit can change that. The convergent-validity claim is therefore conditional on the extraction stage under either set of bases.

12.5 QUANT is also crude, and shares no denominator with SUS

Independence of construction was bought at the price of commensurability. SUS, HEDGE and BREADTH are densities over 14.4M lemmatised, stopword-filtered tokens; QUANT is a density over 26.2M raw tokens, a base roughly 1.8 times larger and differently composed. No two components of the index share a denominator. Score normalisation makes them addable, so the arithmetic of $ESGSI^{\text{ext}}$ is well defined, but it is the score normalisation that makes them comparable rather than anything about the quantities themselves.

The criterion is also noisy in a way we can name but have not corrected. Its **large_numbers** family matches any run of four or more digits with calendar years excluded, which counts page numbers, note references, ISINs, share counts and monetary figures alongside quantified ESG disclosure; it is 6.92 % of all QUANT hits (13,518 matches) and no part of it was validated against ESG content. And the measure has a hard floor that one document reaches: a single report of the 343 records zero QUANT matches across all four families, so its $Z(\text{QUANT})$ is bounded below by the sample and is not a measurement of anything. That document is also among the shortest extractions in the corpus, which is the dependence §9 already notes — QUANT and SUS are computed over the same zones, and a report from which extraction recovered little text is low on both for reasons unrelated to disclosure.

12.6 The vocabulary: a recall floor and its own construction artefacts

Recall is bounded by the vocabulary at both the extraction and counting stages: an unlisted term is not under-weighted but absent. An external audit of 880 zones from four FY2023 reports found 147 ESG terms present in the text and absent from our vocabulary, over 2,970 occurrences, led by *compliance* (436) and *social* (275). None of the 147 is incorporated into the adopted vocabulary, every figure in this paper is computed on the 154 entries, and recall is bounded by that omission. The audit shares the bound it measures: it matched lexically, inside already-extracted zones, so paraphrase is invisible to it too.

A sector-bias analysis has since been run over all 343 reports, and it converts one item on this list from an unresolved omission into a design choice we disclose rather than defend as necessary. Of the audited candidates, 22 are concentrated enough in a single industry to be called sectoral, and they are excluded; they carry 2.0 % of the 143 candidates’ occurrence mass and 0.37 % of the substance numerator they would be added to. The 22 natural-form entries yield 19 live TF-IDF entries: *b corp*, *iso 37001* and *cop 28* do not survive preprocessing, and two of the survivors are lemmatisation artefacts of the class this subsection goes on to describe — *better cotton* becomes

well cotton and *living income* becomes *live income*, the latter scoring two occurrences in one firm. Added on top of the expanded 289-term robustness vocabulary of §10.5, they move mean SUS^{dens} from 6.5162 to 6.5406, the flagged count from 164/343 to 167/343 and the trend slope from -0.194 to $-0.193/\text{yr}$; the two indices correlate 0.99980 and three labels of 343 change.

Some of them do what the objection predicts, and that is why they are out: mean SUS^{dens} rises 0.354 for Danone (*nutrition, food safety*), 0.111 for Enel (*energy storage*) and 0.099 for Schneider Electric, against a corpus mean of 0.024, and 34 of the 49 firms gain less than 0.02 — firms credited for the industry they operate in rather than for what they disclosed. The exclusion avoids a bias we can name and measure, and we prefer it on that ground; what it does not do is carry any result.

The bin is not one kind of term. The rule sorts on concentration without distinguishing its causes, so alongside industry vocabulary it holds house-style governance language — *executive compensation* appears in 29 of the 49 companies and is excluded on the second limb of the rule, that half its occurrences sit in one firm — and one conference name. Two entries were measured as zero by a tokeniser that split on letters only: re-measured with word boundaries, *sf6* is correctly sectoral and *cop28* meets the acceptance rule and sits in the sectoral file in error. And the dispersion of *board oversight* depends on the base: 8 companies as a natural-form phrase, 30 as the lemmatised bigram the vocabulary actually matches, because stopword removal manufactures the adjacency. The rule was applied consistently; its inputs were not all measured on the representation the instrument scores.

Against the 96 labels of 343 that vector normalisation moves (§10), the 22 sectoral terms move three, so no claim in this paper would read differently had we included them. The decision is disclosed and immaterial, not necessary; the note in `esg_terms.txt` deferring it to a sector-bias analysis overstated what such an analysis could settle.

Precision is bounded too, by the construction rule §5 defends. Passing every vocabulary entry through the corpus preprocessor guarantees that vocabulary and text share a representation, and it produces entries that are lemmatisation artefacts rather than terms: *esr* from *esrs*, *whistleblowe* from *whistleblowing*, *ecovadi*, *sustainalytic*, *live wage*, *employee wellbee*, *datum breach*. The rule makes these harmless for matching, since the corpus passes through the same transformation, but not harmless for weighting. *esr* occurs 6,282 times across 74 of the 343 documents at $\text{idf} = 2.52$, placing it in the high-idf group that §7 identifies as the one the idf specifications most heavily up-weight; it is the largest single entry in that group. The rare-term critique we direct at $SUS^{\text{tfidf-L2}}$ therefore applies to our own vocabulary, and it applies to $SUS^{\text{tfidf-L}}$ as well as to $SUS^{\text{tfidf-L2}}$. Only SUS^{dens} , which weights every entry equally, is insensitive to it. We did not audit the vocabulary for artefacts before running the analysis, and no entry was removed on that ground.

12.7 One extraction pipeline, one tone dictionary

Every result is conditional on the zone extraction of §4, whose free parameters — a density threshold of 1.0 keyword per 100 words, a ± 1 -paragraph context window, a 150-character minimum zone — were set by judgement and never varied. That stage discards roughly two thirds of each report and sits upstream of every number here, and it has no counterpart in the original, so the limitation is ours alone.

The extraction stage shares its vocabulary with the measurement stage, so building the robustness vocabulary of §10.5 varied it incidentally, and the effect can be read off. Holding the adopted 154-term scoring vocabulary fixed and swapping only the extraction — 154 natural-form terms against 296 — takes the corpus from 14.4 to 16.4 million lemmatised tokens, lowers mean SUS^{dens} from 5.897 to 5.168, moves 7 of 343 labels and leaves the two indices correlated at 0.988. The added zones are therefore poorer in adopted-vocabulary terms than the zones already admitted, which is what the density threshold predicts of text entering at the extensive margin. This is one point, not a sensitivity analysis: the three free parameters of §4 — the density threshold, the context window and the minimum zone length — are still never varied.

SEN is a dictionary count and inherits the known limitations of Loughran and McDonald (2011) on non-financial narrative text: no negation or intensifier handling, a 140-positive against 893-negative asymmetry in the stem lists, and a bounded polarity ratio that is insensitive to the volume of matched terms — a report with 4 positive and 2 negative matches scores exactly as one with 400 and 200. Our input is lemmatised text that the LM tokeniser then stems again. Substituting this dictionary for the original’s is itself a divergence (§1) and one more reason the two indices are not the same instrument — as is our vocabulary, which cannot be aligned with the original’s because the original’s is unpublished (§5).

12.8 No external validation, and no blind specification choice

The check in §9 is a construct check against an internal yardstick, not a criterion check. Neither this study nor the original tests ESGSI against an external greenwashing benchmark — regulatory findings, controversy data, or hand-labelled cases. We identify that absence in the original (§2) and concede it here in the same terms: ESGSI is a measure whose validity rests on its definition. The concession runs further than the original’s does. Our one positive methodological claim — that SUS^{dens} is the specification to prefer — is decided by a criterion we built ourselves, from our own regexes over our own extracted zones. The original’s specification is undisclosed and ours is disclosed, but neither is anchored to anything outside its author’s pipeline.

Nor was the comparison blind. The vocabulary, the extraction parameters, the QUANT regex families and the specification set were all fixed by us with the results visible, and no part of the corpus was held out. Every individual choice is disclosed in §A and each is defended where it is made, which is the disclosure standard we hold the original to; disclosure is not the same as pre-commitment, and we claim only the first. The classification thresholds of §10.5 — twelve companies, half the occurrences in one firm — were set later than the rest and with the results they adjudicate already computable, so they fall inside this concession rather than outside it.

A Reproducibility

This appendix is the deliberate counterpart to §2: where that section catalogues what Lagasio (2024) leaves unspecified, this one discloses every corresponding element of our own pipeline at the same level of detail, because a paper that audits another study’s reproducibility and then reports its own loosely would be self-refuting.

A.1 Pipeline and intermediate artefacts

The pipeline is a fixed sequence of four stages, each reading the previous stage’s output file and writing its own; there is no step that reads back from a later stage. Table 29 gives the module, input and output artefact of each. The whole sequence is invoked as `python main.py`, with three boolean flags (`run_extraction`, `run_preproc`, `run_esgsi_analysis`) all set to `True` and a fourth (`run_lda`) set to `False`; §A.3 explains why.

A.2 Dependencies

`pyproject.toml` declares dependency names without version constraints, so it is not itself a reproducibility record. Table 30 gives the resolved environment (conda environment `esgwashing`) actually used to produce `results.csv`, which is the binding version record for this paper.

A.3 Determinism

Extraction, preprocessing and measurement together are a pure function of the PDFs and the lexicon files in Table 29: no network call, no external API, and no random seed anywhere on this

Table 28: Attribution of each limitation. “Inherited” means the property follows from the index architecture of Lagasio (2024) and holds for the original as well.

Limitation	Origin
Zero threshold is definitional, not calibrated	Inherited
No cardinal scale; scores are sample-relative	Inherited
idf corpus dependence of $SUS^{\text{tfidf-}\ell}$, $SUS^{\text{tfidf-}L2}$	Inherited
Vocabulary not comparable to the original’s	Inherited (list unpublished)
No external greenwashing benchmark	Shared
Country / document-type confound	Ours (corpus design)
Cross-section of 49 firms; no clustered inference	Ours
ESGSI ^{ext} weights $w_q = w_h = 0.5$	Ours
Criterion QUANT shares 14 terms with SUS (7.50 % of SUS mass); levels in §9 are upper bounds	Ours
Criterion and measure read the same extracted zones; not removable	Ours
No component shares a token base with any other	Ours
large_numbers counts non-ESG figures (6.92 % of QUANT)	Ours
QUANT floors at zero for 1 of 343 documents	Ours
Lexical recall floor (147 audited omissions)	Ours
Sectoral vocabulary excluded by choice (moves 3 labels of 343)	Ours (disclosed)
Lemmatization artefacts up-weighted by idf (<i>esr</i>)	Ours
Single untested extraction stage	Ours (no counterpart)
Extraction vocabulary varied once, incidentally (7 labels of 343)	Ours
SEN dictionary limitations	Ours (dictionary substituted)
Specification chosen non-blind, no held-out data	Ours

Table 29: Pipeline stages and intermediate artefacts.

Stage	Module	Input	Output artefact
Extraction	src/lexical_document_filter.py (LexicalDocumentFilter)	data/pdf/*.pdf (344 files)	data/chunks_lexical/*.json (344 files, one per PDF)
Preprocessing	src/text_processor.py (TextProcessor)	relevant_text field of each JSON	data/clean/processed_texts.csv (;-separated; clean_text, raw_text columns)
Lexicon build (run once, offline)	scripts/build_lexicons.py	metadata/esg_terms.txt, metadata/RAW_LM_dictionary	metadata/esg_terms_lemmatized.txt, metadata/lm_hedge_lemmatized.txt (1,263 forms)
Measurement & index assembly	src/esgsi_analyzer.py (ESGSIAnalyzer), after an MD5 content-hash dedup on clean_text inside main.py	processed_texts.csv (344 rows) + the two generated lexicons + metadata/muestras_informes.xlsx	results/metrics/results.csv (343 rows, 17 columns)

Table 30: Runtime dependencies and versions.

Package	Version
Python	3.11.15
pandas	3.0.1
scikit-learn	1.8.0
spaCy	3.8.11
en_core_web_md	3.8.0
PyMuPDF	1.27.2
gensim	4.4.0
pysentiment2	0.1.1
loguru	0.7.3
openpyxl	3.1.5

path. The one non-trivial control-flow step, the deduplication that drops `results.csv` from 344 to 343 rows, is itself an MD5 hash of `clean_text` with the first occurrence kept and the second discarded (§3) — a deterministic function of the corpus, not a random draw. The predecessor pipeline queried a hosted embedding model for its semantic filter (§4); that dependency was removed in its entirety. The only sampling anywhere in the codebase is a draw of 20 PDFs (seed 7) used to estimate extraction yield as a share of raw PDF text (§4); it feeds a descriptive statistic, not any column of `results.csv`. Unlike every other quantity in this determinism claim, this one draw is not backed by a script in the repository: no committed file performs it, so it cannot be independently re-run from the code as it stands, only checked against the seed value and resulting statistic recorded here and in §4. We record this rather than imply a reproducibility this step does not have. The only stochastic component in the repository is the LDA topic model (`src/topic_modeler.py`): `LdaModel.fit()` is called with `random_state=None` and is therefore unseeded on every run that would produce coherence figures; only the separate stability analysis (`compute_stability`, gated behind `LDA_N_SEEDS>0` and never invoked at the repository’s current setting of `LDA_N_SEEDS=0`) explicitly seeds successive runs 0 through `n_seeds-1`. LDA is switched off in its entirety (`run_lda=False`) and feeds none of SUS, SEN, QUANT, HEDGE, ESGSI or ESGSI^{ext}. Its own output file, `results/lda/coherence.csv`, predates the current code and does not match its column schema, so we exclude it from every claim in this paper rather than cite a stale artefact.

A.4 Index parameters

Table 31 consolidates, in one place, every free parameter used from extraction through index assembly. Each value is defined and justified where the corresponding stage is described (§4, §5, §6); nothing here is derived independently of those sections. The index-stage rows repeat Table 11 exactly, which is the definitional source for them; this table adds the extraction and preprocessing stages and resolves each value to the identifier the code uses. Where the two tables would disagree, §6 governs and this one is in error. *Z* denotes score normalisation throughout; it is never vector normalisation (§6).

A.5 Output schema

Which artefacts correspond to which specification

The repository carries three complete runs, and the distinction matters because they were produced by different vocabularies and, since the vocabulary also drives paragraph selection (§10.5), by different extractions. `results/metrics_baseline_paper/` holds the adopted 154-term specification and is the source of every figure reported in this paper. `results/metrics/` and `results/metrics_sec/` hold the 289-term and 289 + 22-term robustness runs of §10.5. The working tree’s

Table 31: Parameters, extraction through index assembly.

Stage	Parameter	Value
Extraction	Keyword-density threshold	1.0 keyword / 100 words
Extraction	Context window	± 1 paragraph
Extraction	Minimum zone length	150 characters
Extraction	Margin bands	top 8% / bottom 8% of page height
Extraction	Header/footer repeat threshold	$> 30\%$ of pages, candidate blocks ≤ 120 characters
Extraction	Financial-table digit rule	digit share > 0.35 , ≤ 6 lines, no ESG keyword present
Extraction	Financial-table line rule	every line < 60 characters, ≥ 2 lines, first line without terminal punctuation, no ESG keyword present
Extraction	Section-header rule	all-caps block, ≥ 2 letters, ≤ 10 tokens, $< 60\%$ single-character tokens, no dot-extension token (excludes filenames)
Preprocessing	spaCy pipeline	<code>en_core_web_md</code> , <code>ner</code> and <code>parser</code> disabled
Preprocessing	Token filter	alphanumeric, ≥ 2 alphabetic characters, length > 2 , not stop/punct/space, lemma $\notin$ 189-entry stopword list
Preprocessing	<code>nlp.max_length</code>	5,000,000 characters
SUS (SUS ^{tfidf-L2})	TfidfVectorizer + aggregation	<code>vocabulary=154</code> lemmatised terms; <code>ngram_range=(1,3)</code> ; <code>norm='l2'</code> ; <code>smooth_idf=True</code> ; <code>sublinear_tf=False</code> ; document score is the <i>mean</i> of the L2-normalised term weights over the 154-term vocabulary (<code>.mean(axis=1)</code>)
SUS (SUS ^{dens} , SUS ^{tfidf-ℓ})	Denominator + aggregation	lemmatised token count L_d ; document score is the <i>sum</i> of (idf-weighted, for SUS ^{tfidf-ℓ}) counts over the vocabulary, divided by L_d , per-100-word density; no vector normalisation
SEN	Dictionary + smoothing	pysentiment2 bundled Loughran-McDonald list (Loughran and McDonald, 2011), Porter-stemmed at load (140 positive / 893 negative stems); polarity ratio with $\varepsilon = 10^{-6}$
QUANT	Regex families	percentages; large numbers (≥ 4 digits, calendar years excluded); units; framework acronyms; density over raw-text tokens R_d
HEDGE	Lexicon	1,263-form lemmatised L&M list (Uncertainty, WeakModal, StrongModal, Constraining); density over L_d
Z (score norm.)	Estimator	population sd, ddof = 0; zero vector if sd = 0
ESGSI ^{ext}	Weights	$w_q = w_h = 0.5$ (<code>config.ESGSI_EXT_WEIGHTS</code> , arbitrary)
Labelling	Threshold	ESGSI $> 0 \Rightarrow$ "Potential ESG-washing"; else "Likely Genuine"

`metadata/esg_terms_lemmatized.txt` is the 289-entry expansion; the adopted 154-entry vocabulary is recoverable from the repository history at commit `ce6804a`, and `metadata/esg_terms_sectorial.txt` holds the 22 terms the classification set aside. A reader reproducing the paper’s numbers needs the 154-entry list, not the working-tree file.

Each `results.csv` is ;-separated, 343 rows by 17 columns, listed in Table 32. The row count is already net of the single duplicated report that deduplication removes (§3); nothing downstream needs to drop a row by hand. Because QUANT and HEDGE have small standard deviations (0.0035 and 0.0096), recomputing ESGSI^{ext} from the rounded columns reproduces it only to ± 0.009 ; recomputing from the unrounded pipeline values is exact. The ESGSI column carries no such hazard: no document’s index falls within the rounding interval of the labelling threshold, so counting strictly positive rounded ESGSI values returns 171, exactly the number of *Potential ESG-washing* labels in the `Etiqueta` column.

Table 32: `results.csv` column schema.

Column(s)	Content	Rounding
Documento, País, Compañía, Año, TipoDocumento	Filename, country, company, year, document type	none
SUS_Score, SUS_density, SUS_tfidf_length, SUS_lagasio	Three substance specifications, plus the one selected by <code>config.SUS_MODE</code> (SUS_Score, currently SUS ^{dens})	6 dp
Breadth, SEN_Score, QUANT_Score, HEDGE_Score, ESGSI, ESGSI_ext	Remaining components and the two composite indices	4 dp
Etiqueta, Etiqueta_ext	Binary label from ESGSI, ESGSI ^{ext}	categorical

A.6 Vocabulary and hedge lexicon as published artefacts

Against Lagasio (2024), whose keyword list is attributed to GRI (2021) and SASB (2018) but never sized, itemised or published, and whose data availability statement reads “No” (§2), every lexicon that feeds a score in this paper is a plain-text file in the repository, one canonical form per line, loaded by `config.py` at run time rather than transcribed into the paper: the 154-term ESG vocabulary (`metadata/esg_terms.txt`, natural singular surface forms across seven thematic blocks) and its derived lemmatised form (`metadata/esg_terms_lemmatized.txt`, 85 unigrams, 62 bigrams, 7 trigrams) that the vectoriser actually consumes; the 1,263-form lemmatised Loughran–McDonald hedge lexicon (`metadata/lm_hedge_lemmatized.txt`), built from the Uncertainty/WeakModal/StrongModal/Constraining subset (1,245 of 9,752 rows) of the L&M dictionary (Loughran and McDonald, 2011); and the 189-entry supplementary stopword list (`metadata/personal_stopwords.txt`) that participates in preprocessing. The files a third party would need to recompute every number in this paper are the same files that produced them.

A.7 Four silent-failure defects, repaired

The four defects below were found and fixed together in one commit, `44114d0` (*fix: repair ESG lexicon pipeline and restore audited vocabulary*), but they do not share one origin. Two of them – the QUANT routing and the HEDGE lemmatisation mismatch – were introduced when `calculate_quant_scores` and `calculate_hedge_scores` were first written, in the same commit that retired the semantic filter for the current deterministic pipeline (`64393f2`, §4). The other two predate that migration entirely: the alphanumeric token filter traces to `f8bd8a5` (2026-03-21) and the vectoriser’s default `ngram_range=(1,1)` to `fbcab9f` (2026-03-20), both about a

week before the semantic filter was even added (813005c, 2026-03-28) and five weeks before it was retired in 64393f2. All four therefore survived not one refactor but the full semantic-filter era and its retirement – more than four months and several intervening commits – undetected. Every figure in Table 33 is measured on the corpus rebuilt after 44114d0. We report them as a methods contribution rather than as housekeeping: each produced a plausible-looking number instead of an error, and that failure mode is the one most likely to survive peer review in dictionary-based text measurement.

A provenance caveat belongs here rather than left implicit. 44114d0 is HEAD on this branch, and every commit-level claim above is checked directly against it. It is not, on its own, sufficient to regenerate `results.csv`: the three-specification SUS system (SUS^{dens} , $SUS^{\text{tfidf-}\ell}$, $SUS^{\text{tfidf-L2}}$ and the `config.SUS_MODE` switch between them), the `Breadth` column, and the deduplication step of §A.3 exist only as uncommitted changes to `main.py`, `src/config.py` and `src/esgsi_analyzer.py` on top of that commit. A reader who checks out 44114d0 exactly gets a single, non-selectable TF-IDF-mean SUS score, no `Breadth` column, and 344 unduplicated rows – not the pipeline this paper describes. Closing this gap requires a further commit capturing that working-tree state; until it exists, the commit hash names a necessary but not sufficient condition for reproduction.

Table 33: Silent-failure defects found and repaired in 44114d0.

Component	Silent failure	Fix
QUANT	<code>percentages/large_numbers</code> ran on lemmatised <code>clean_text</code> , which preprocessing had already stripped of digit-bearing tokens; both families matched 0 occurrences in all 344 documents	Routed onto the untouched <code>raw_text</code> column; post-fix, 109,974 and 13,518 matches respectively
SUS vocabulary	<code>TextProcessor</code> kept only <code>token.is_alpha</code> tokens, silently deleting alphanumeric ESG terms (<code>co2</code> , <code>co2e</code> , <code>sf6</code> , <code>iso14001</code>) from <code>clean_text</code> wherever they occurred, so the corresponding vocabulary entries could never match downstream	Filter replaced by an alphanumeric test requiring ≥ 2 alphabetic characters (still excludes pure figures and page references); <code>co2</code> and <code>co2e</code> occur 13,376 and 3,390 times in the corrected corpus
HEDGE	The L&M dictionary is distributed in inflected form; matched directly against lemmatised <code>clean_text</code> , it covered only $\approx 46\%$ of its own entries	Generated lexicon is the union of each original form and its spaCy lemma: 1,196 unique inflected forms $\rightarrow$ 1,263 forms
SUS (multi-word entries)	<code>TfidfVectorizer/CountVectorizer</code> used sklearn’s default <code>ngram_range=(1,1)</code> ; every bigram or trigram vocabulary entry (69 of 154, 45%) scored zero in every document, with no warning raised	<code>max_n</code> derived at runtime from the vocabulary itself (<code>max(len(k.split())) for k in keywords</code>) = 3 and passed as <code>ngram_range=(1,3)</code>

The general lesson is not specific to this codebase. In dictionary-based measurement, a mis-specified component returns zero rather than raising an error: a token filter, a lemmatisation mismatch or a vectoriser default can each silently remove an entire category of matches while the pipeline runs to completion and writes a number that looks exactly like a low-incidence finding. The absence of a signal is therefore indistinguishable, from the output alone, from the absence of a defect. None of the four defects in Table 33 raised an exception, a warning, or an out-of-range value; each was found only by checking a component against text known in advance to contain

the pattern it was supposed to detect. Components in this pipeline must therefore be, and were, validated individually against known-positive text before their output is trusted in any of the results that follow.

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